

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

Chap. 4 Open-Loop Discrete-Time Systems

Chap. 5 Closed-Loop Systems

Chap. 6 System Time-Response Characteristics

Chap. 7 Stability Analysis Techniques

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6.2 System Time Response

Example 6.1

The unit-step response will be found for the first-order system in Figure 6-1a. Since the plant of a temperature control system is often modeled as a first-order system, this system might then be the model of a temperature control system (see Section 1.6). Using the techniques developed in Chapter 5, we can express the system output as

$$C(z) = \frac{G(z)}{1 + G(z)} R(z)$$

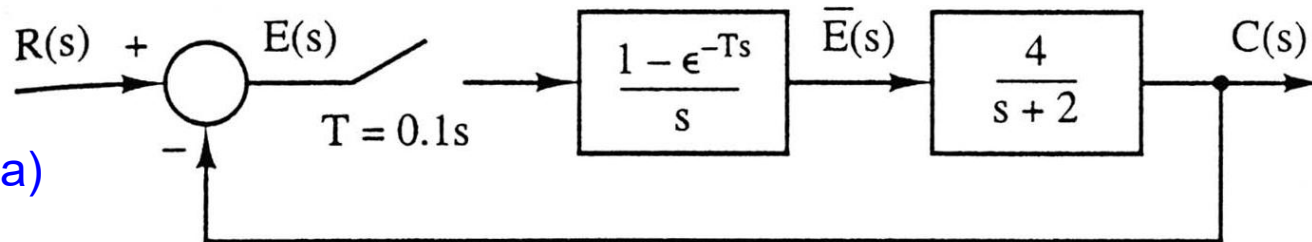


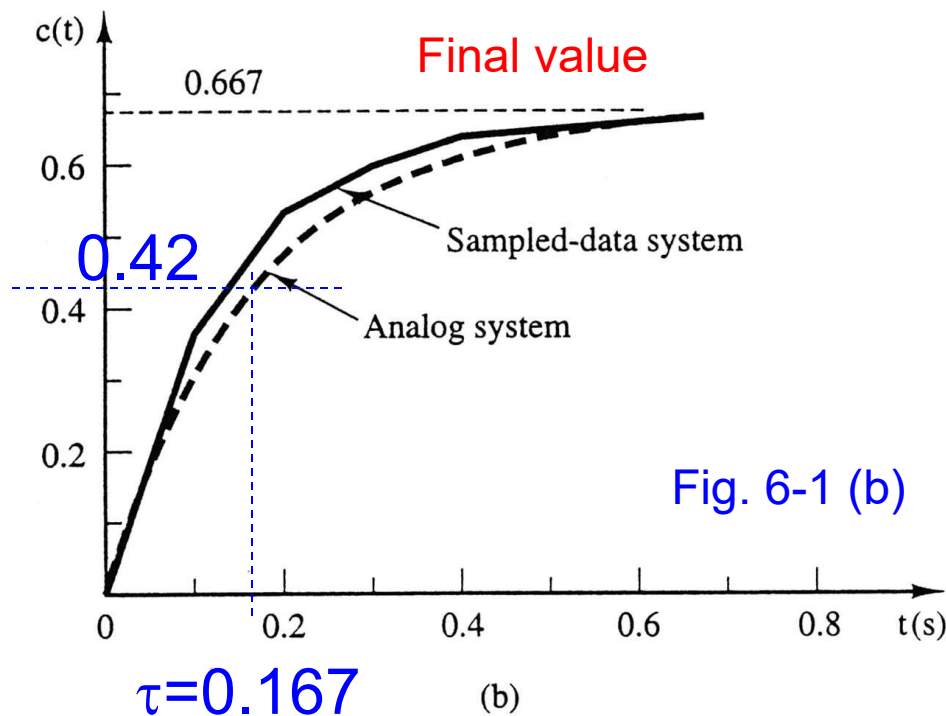
Fig. 6-1 (a)

where $G(z)$ is defined as

$$\begin{aligned} G(z) &= \mathcal{Z} \left[\frac{1 - e^{-Ts}}{s} \frac{4}{s + 2} \right] = \frac{z - 1}{z} \mathcal{Z} \left[\frac{4}{s(s + 2)} \right] \\ &= \frac{z - 1}{z} \frac{2(1 - e^{-2T})z}{(z - 1)(z - e^{-2T})} = \frac{0.3625}{z - 0.8187}, \quad T = 0.1 \text{ s} \end{aligned}$$

from the transform table in [Appendix VIII](#). Thus the closed-loop transfer function $T(z)$ is given by

$$T(z) = \frac{G(z)}{1 + G(z)} = \frac{0.3625}{z - 0.4562}$$



Since $R(z) = \mathcal{Z}[1/s] = z/(z - 1)$,

$$C(z) = \frac{0.3625z}{(z - 1)(z - 0.4562)} = \frac{0.667z}{z - 1} + \frac{-0.667z}{z - 0.4562}$$

The inverse z -transform of this function yields the system time response at the sampling instants. Thus

$$c(kT) = 0.667[1 - (0.4562)^k]$$

This response is listed in Table 6-1. It is seen that the response reaches a steady-state value of 0.667.

TABLE 6-1
RESPONSE
EXAMPLE

kT	$c(kT)$
0	0
0.1	0.36

Example 6.2: Analog System

To show the effect of sampling.

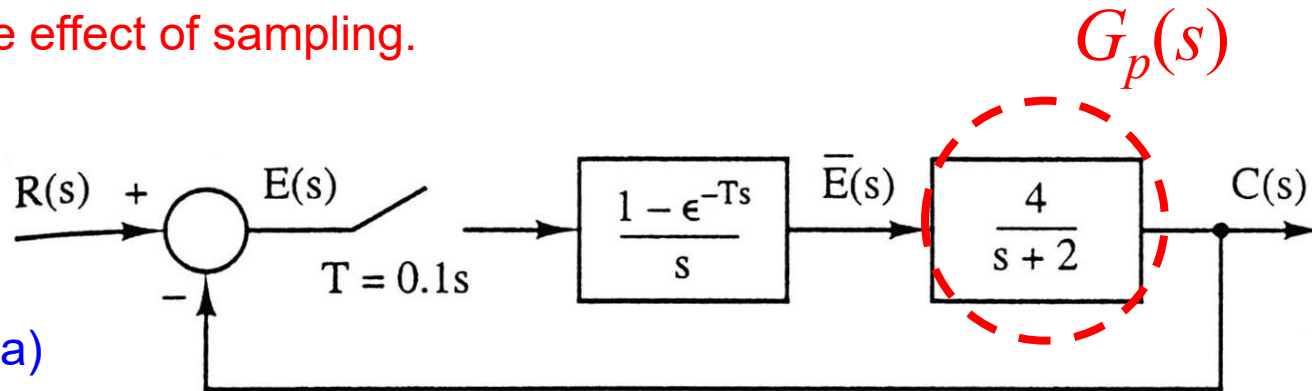


Fig. 6-1 (a)

$$T_a(s) = \frac{G_p(s)}{1 + G_p(s)} = \frac{4}{s + 6}$$

$$C_a(s) = \frac{4}{s(s + 6)} = \frac{0.667}{s} + \frac{-0.667}{s + 6}$$

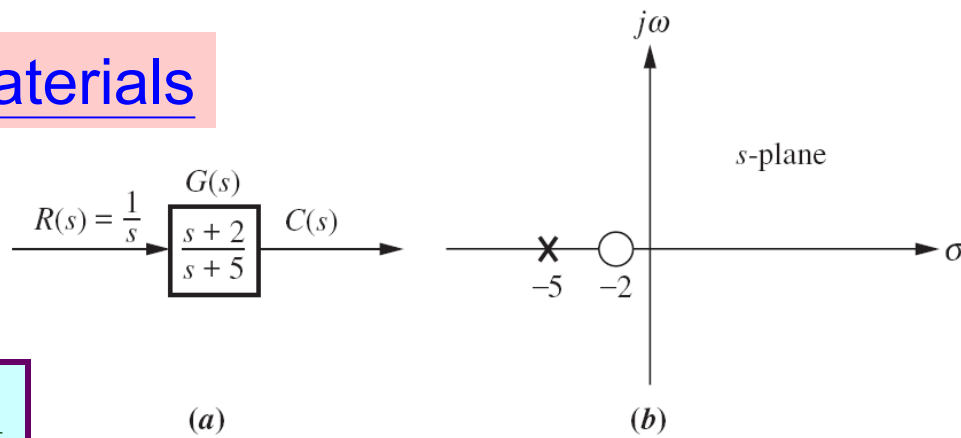
$$c_a(t) = 0.667(1 - e^{-6t})$$

↓ Forced/
Steady-state resp.
 ↓ Natural/
Transient resp.

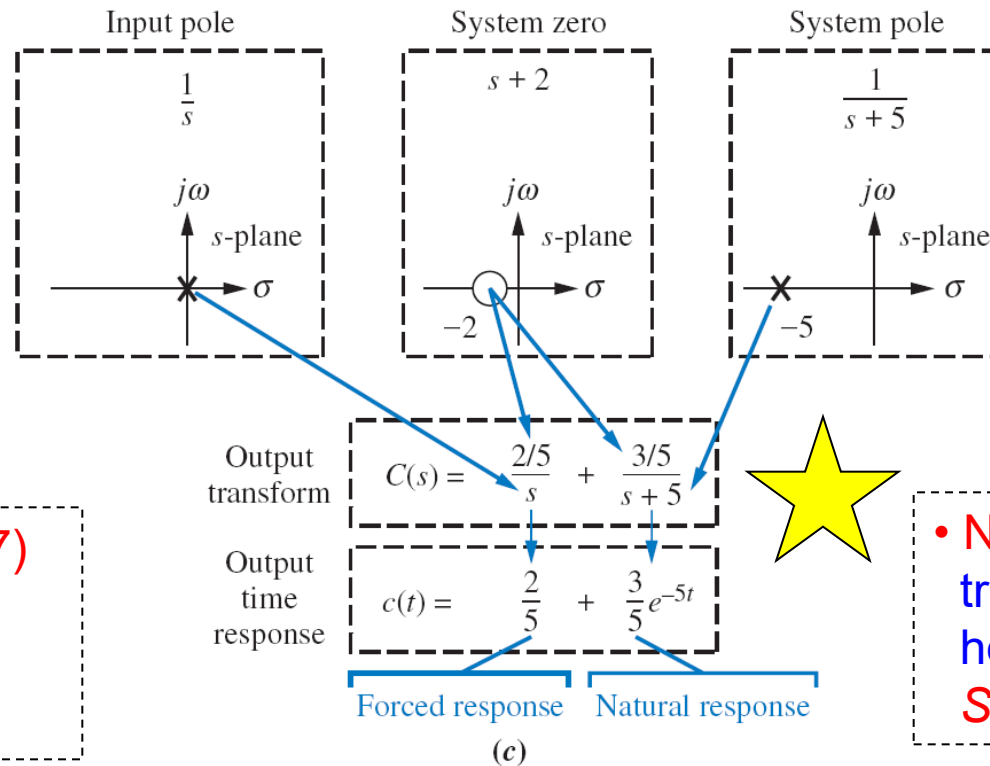
TABLE 6-1
RESPONSES FOR
EXAMPLE 6.1

kT	$c(kT)$	$c_a(t)$
0	0	0
0.1	0.363	0.300
0.2	0.528	0.466
0.3	0.603	0.557
0.4	0.639	0.606
0.5	0.654	0.634
0.6	0.661	0.648
⋮		
1.0	0.666	0.665

Supplementary Materials



Open-Loop System

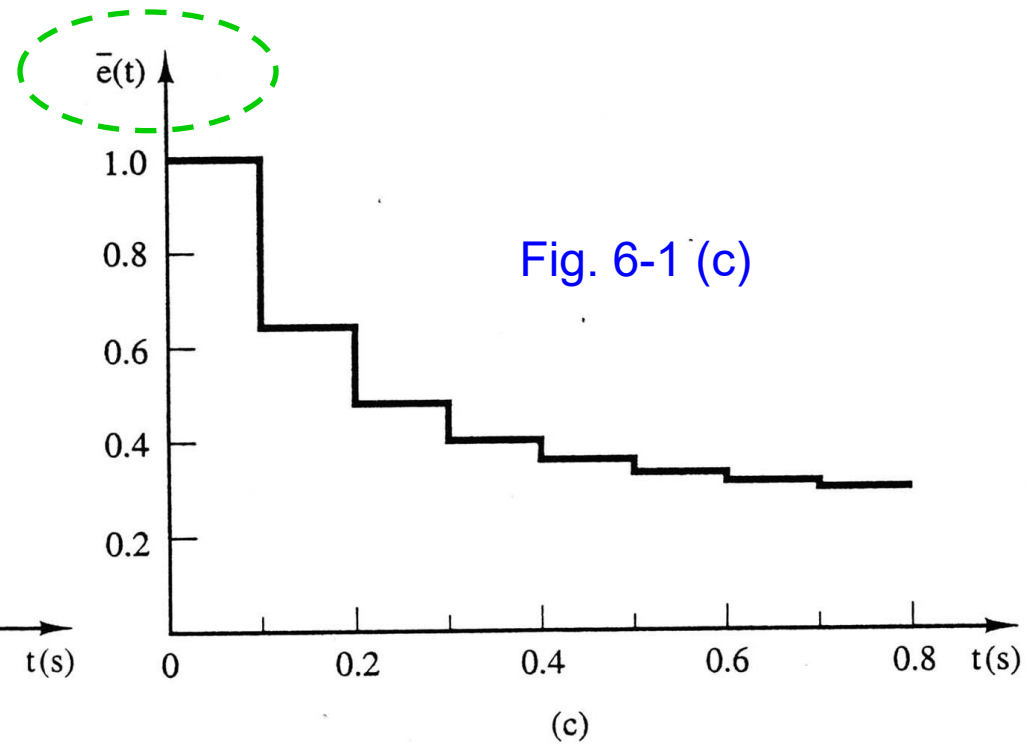
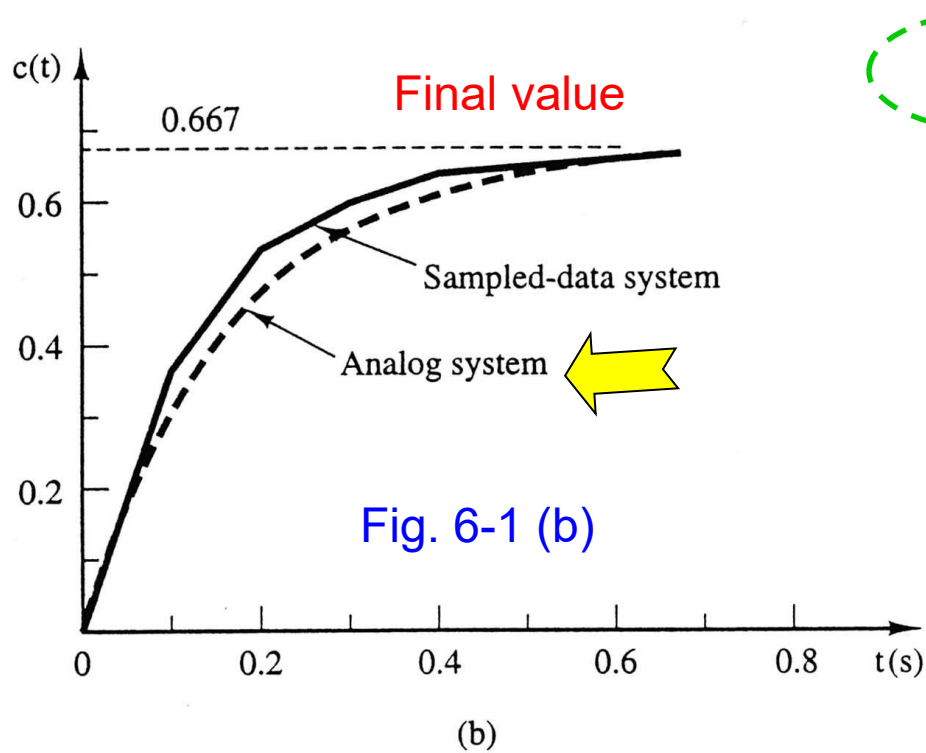


• Force Resp. (Ch-7)
steady-state resp.
particular sol.
Input

• Nat. Resp. (Ch-4)
transient resp.
homogenous sol.
System

FIGURE 4.1 a. System showing input and output; b. pole-zero plot of the system; c. evolution of a system response. Follow blue arrows to see the evolution of the response component generated by the pole or zero

Example 6.2



4.3 FIRST-ORDER SYSTEMS

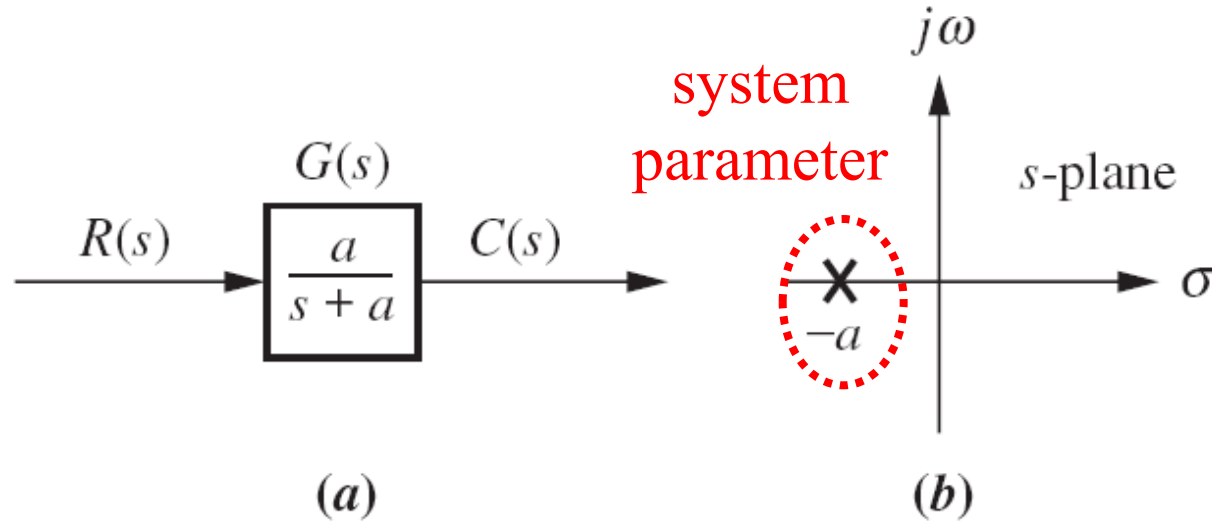
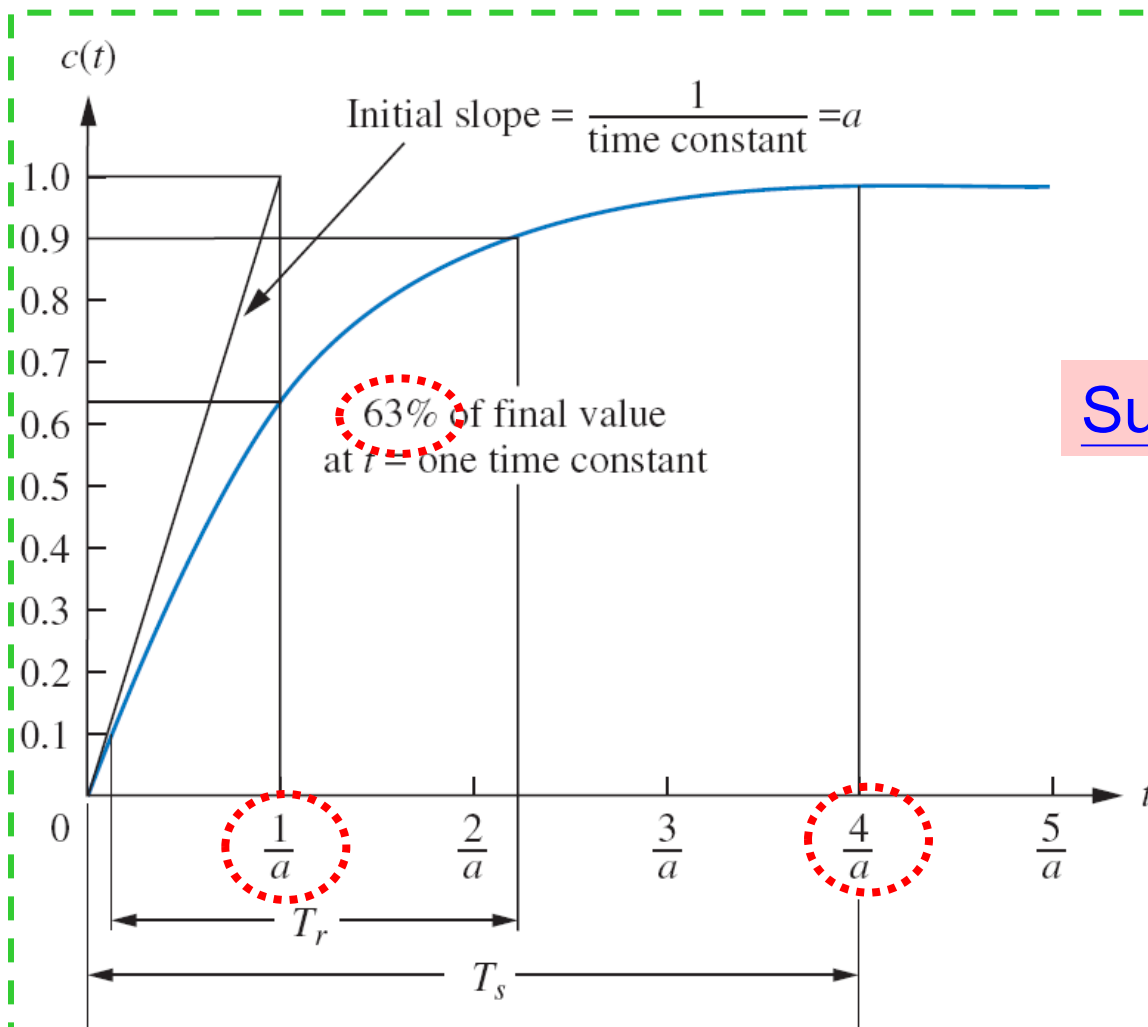


FIGURE 4.4 a. First-order system;
b. pole plot

$$C(s) = R(s)G(s) = \frac{a}{s(s+a)} \quad (4.5)$$

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at} \quad (4.6)$$

FIGURE 4.5 First-order system response to a unit step



[Supplementary Materials](#)

When $t = 1/a$

$$e^{-at}|_{t=1/a} = e^{-1} = 0.37 \quad (4.7)$$

$$c(t)|_{t=1/a} = 1 - e^{-at}|_{t=1/a} = 1 - 0.37 = 0.63 \quad (4.8)$$

Supplementary Materials

- TIME CONSTANT, $T_c = 1/a$

描述一階LTI系統中，對隨時間變化的輸入信號的響應能力的參數。
值愈小，其響應變化就愈快。

- RISE TIME, T_r

$$T_r = \frac{2.31}{a} - \frac{0.11}{a} = \frac{2.2}{a} \quad (4.9)$$

- SETTLING TIME, T_s

$$T_s = \frac{4}{a} \quad (4.10)$$

電容器充電 Charging a capacitor

time constant = $R \times C$ where:

time constant is in seconds (s)

R = resistance in ohms (Ω)

C = capacitance in farads (F)

For example:

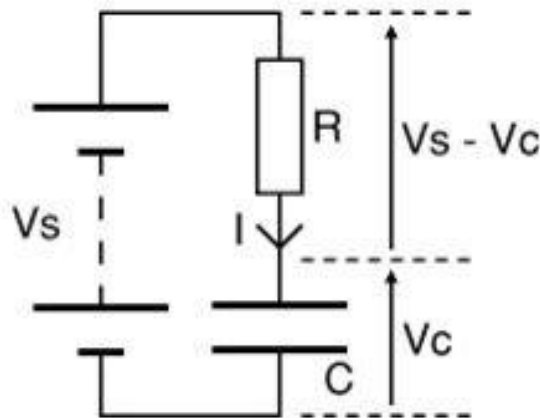
If $R = 47\text{k}\Omega$ and $C = 22\mu\text{F}$, then the time constant, $RC = 47\text{k}\Omega \times 22\mu\text{F} = 1.0\text{s}$.

If $R = 33\text{k}\Omega$ and $C = 1\mu\text{F}$, then the time constant, $RC = 33\text{k}\Omega \times 1\mu\text{F} = 33\text{ms}$.

T_c

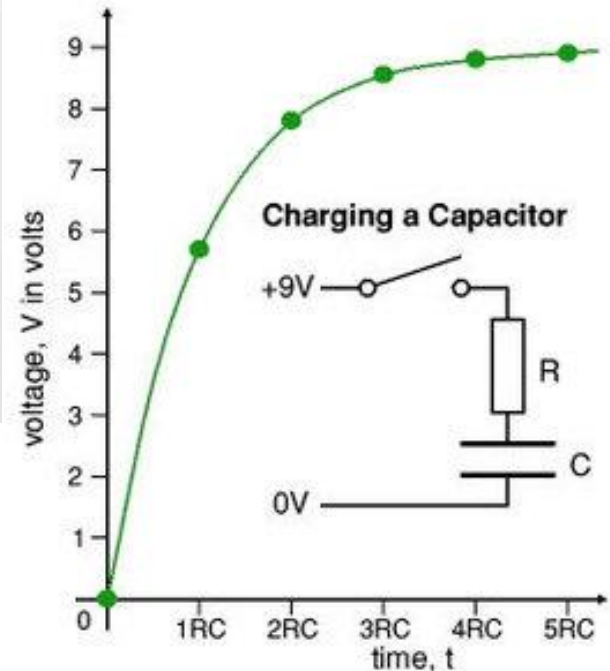
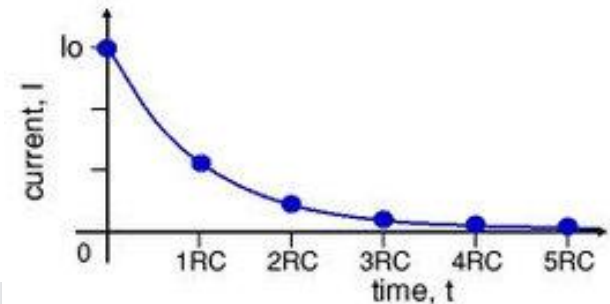
T_s
 $= 4T_c$

Time	Voltage	Charge
0RC	0.0V	0%
1RC	5.7V	63%
2RC	7.8V	86%
3RC	8.6V	95%
4RC	8.8V	98%
5RC	8.9V	99%



Graphs showing the current and voltage for a capacitor charging

time constant = RC



時間常數大，表示電容器充電慢。
注意，時間常數是含有電容電阻之電路的特性，它不是電容器本身的特性。

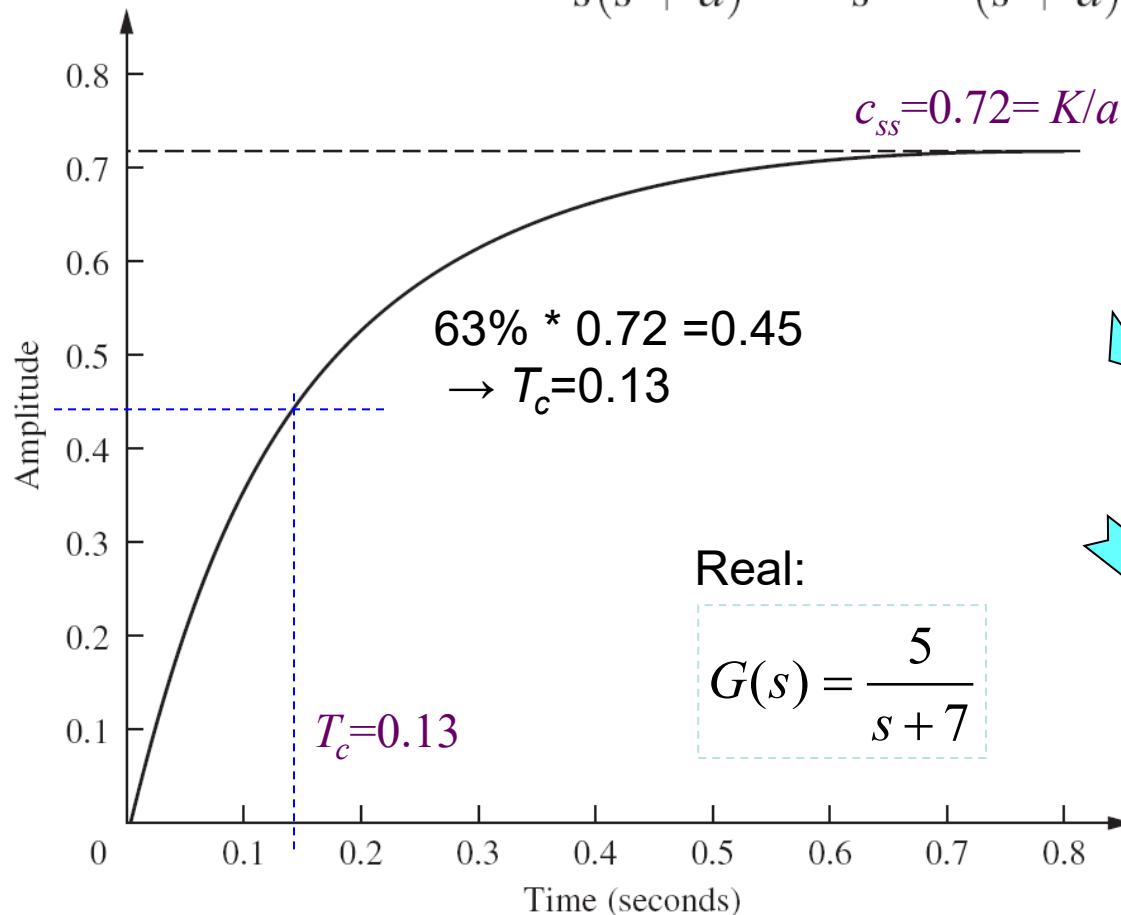
Source: [Link](#)

Supplementary Materials

FIRST-ORDER TRANSFER FUNCTION VIA TESTING

System parameters: a and K

$$C(s) = \frac{K}{s(s+a)} = \frac{K/a}{s} - \frac{K/a}{(s+a)} \quad (4.11)$$



Modeling

By step input

Model (TF)

$$G(s) = \frac{5.54}{s + 7.7}$$

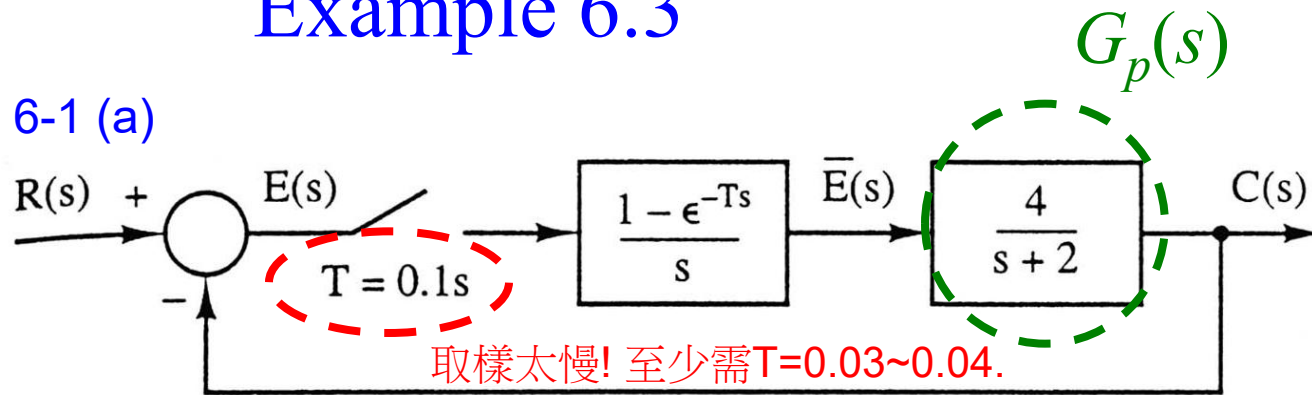
Real:

$$G(s) = \frac{5}{s + 7}$$

FIGURE 4.6 Laboratory results of a system step response test

Example 6.3

Fig. 6-1 (a)



τ : Time constant = 0.167 (ps. $0.667 * 63\% = 0.42$)

A rule of thumb: at least **five samples per time constant**.

Final Value Thm. $\lim_{n \rightarrow \infty} c(nT) = \lim_{z \rightarrow 1} (z-1)C(z) = \frac{G(1)}{1+G(1)} = 0.667$

dc gain $\lim_{z \rightarrow 1} G(z) = \lim_{s \rightarrow 0} G_p(s) \quad (4.14)$

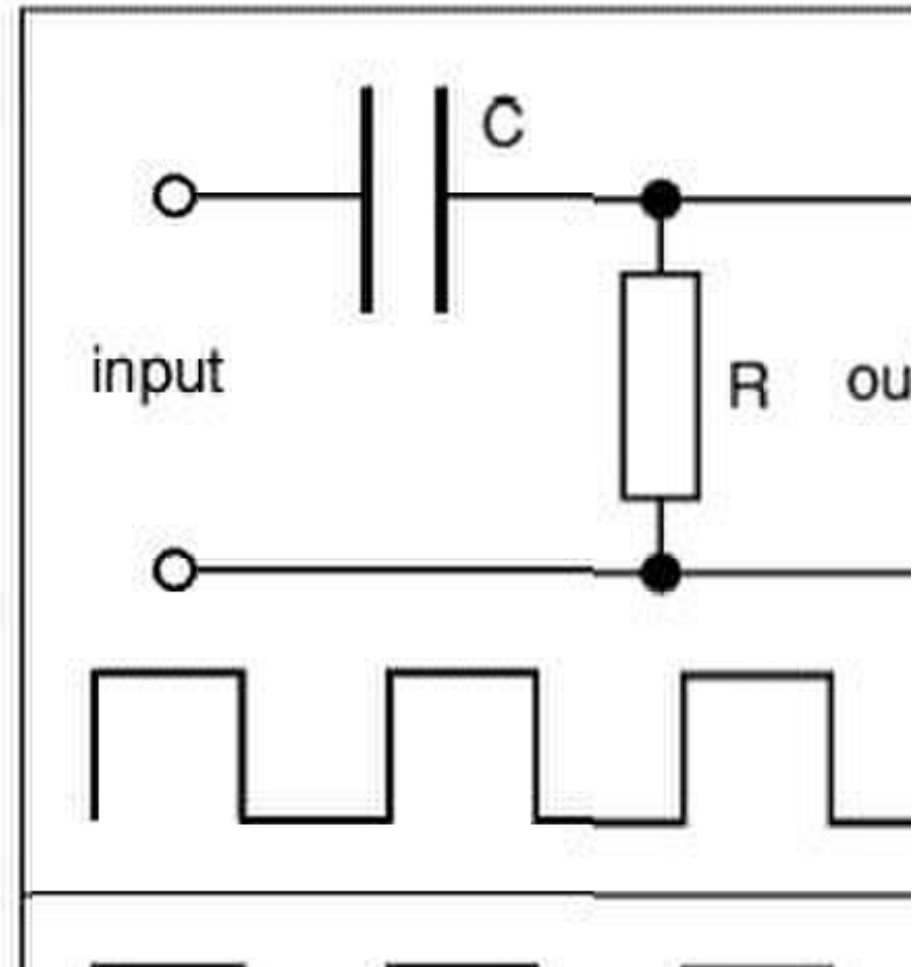
OL dc gain $\lim_{s \rightarrow 0} G_p(s) = 2$

$\Rightarrow$ CL dc gain $\frac{G_p(0)}{1+G_p(0)} = 0.667$

電容耦合 Capacitor coupling

Source: [Link](#)

- 電路的各個部份 (sections) 可能會用電容器彼此相連，因為電容器讓 AC (變化的) 信號通過，而擋下 DC (穩定的) 信號。這稱作電容耦合 (capacitor coupling 或 CR - coupling)。
- 它用在音響系統的級間，讓音響信號 (AC) 通過時，避免任何直流 (DC) 電壓出現其中，例如連接喇叭。它也用作示波器 (oscilloscope) 的「AC」開關。
- 為了在音響系統中成功電容耦合，信號必須以極小或無失真地通過。這可以藉由 RC 常數大於「最低頻信號所需的時間週期 T (典型的 20Hz, $T = 50 \text{ ms}$)」而達成。



A rule of thumb: at least **five samples per time constant**.

2nd-Order System

Example 6.4

The system for this example is shown in Figure 6-2. As in the first example, we will calculate the unit-step response. This system will also appear in many of the following examples. As was demonstrated in Chapter 5, the system output can be expressed as

$$C(z) = \frac{G(z)}{1 + G(z)} R(z)$$

where, from the tables in [Appendix VIII](#)

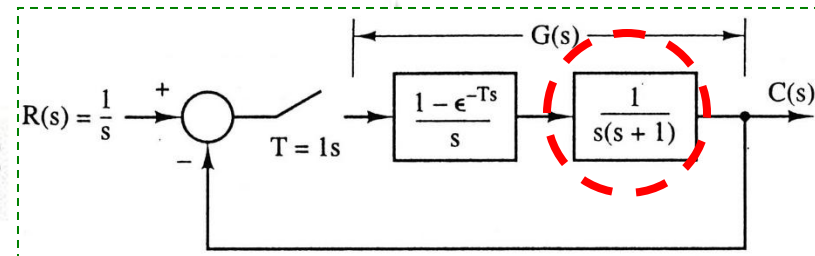
$$G(z) = \left(\frac{z-1}{z} \right) \mathcal{Z} \left[\frac{1}{s^2(s+1)} \right]_{T=1} = \frac{z-1}{z} \left[\frac{z[(1-1+\epsilon^{-1})z + (1-\epsilon^{-1}-\epsilon^{-1})]}{(z-1)^2(z-\epsilon^{-1})} \right]$$

$$= \frac{0.368z + 0.264}{z^2 - 1.368z + 0.368}$$

Then

$$\frac{G(z)}{1 + G(z)} = \frac{0.368z + 0.264}{z^2 - z + 0.632}$$

Since



• **Approach-1: Z-Transform** $R(z) = \frac{z}{z-1}$

then

Z⁻¹: Use Power Series

$$C(z) = \frac{z(0.368z + 0.264)}{(z-1)(z^2 - z + 0.632)} = \underline{0.368z^{-1} + 1.00z^{-2} + 1.40z^{-3}}$$

$$+ 1.40z^{-4} + 1.15z^{-5} + 0.90z^{-6} + 0.80z^{-7} + 0.87z^{-8}$$

$$+ 0.99z^{-9} + 1.08z^{-10} + 1.08z^{-11} + 1.00z^{-12} + 0.98z^{-13}$$

$$+ \dots$$

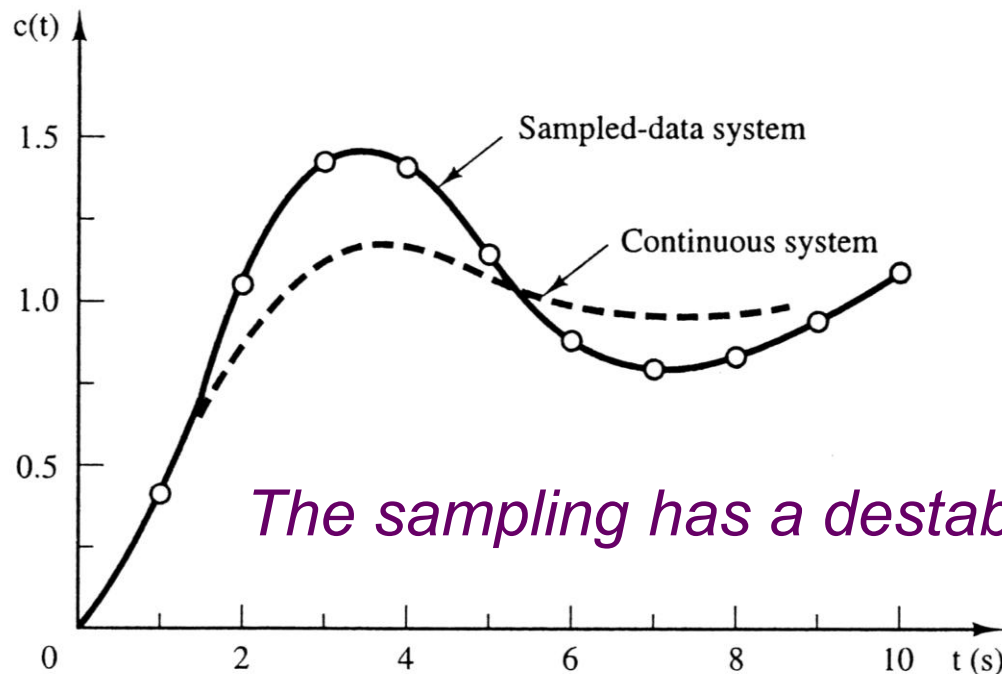
(6-1)

The final value of $c(nT)$, obtained using the final-value theorem, is

$$\lim_{n \rightarrow \infty} c(nT) = \lim_{z \rightarrow 1} (z - 1)C(z) = \frac{0.632}{0.632} = 1$$

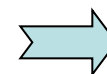
The step response for this system is plotted in Figure 6-3. The response between sampling instants was obtained from a simulation of the system, and, of course, the response at the sampling instants is given in (6-1). Also plotted in Figure 6-3 is the response of the system with the sampler and data hold removed. For this continuous-time system, the transfer function is, in standard notation [1],

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s^2 + s + 1}$$



$$\omega_n = 1$$

$$\zeta = 0.5$$



$$\%OS = 18\%$$

The sampling has a destabilizing effect.

Figure 6-3 Step response of the systems analyzed in Example 6.4.

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

(4.38)

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

(4.39)

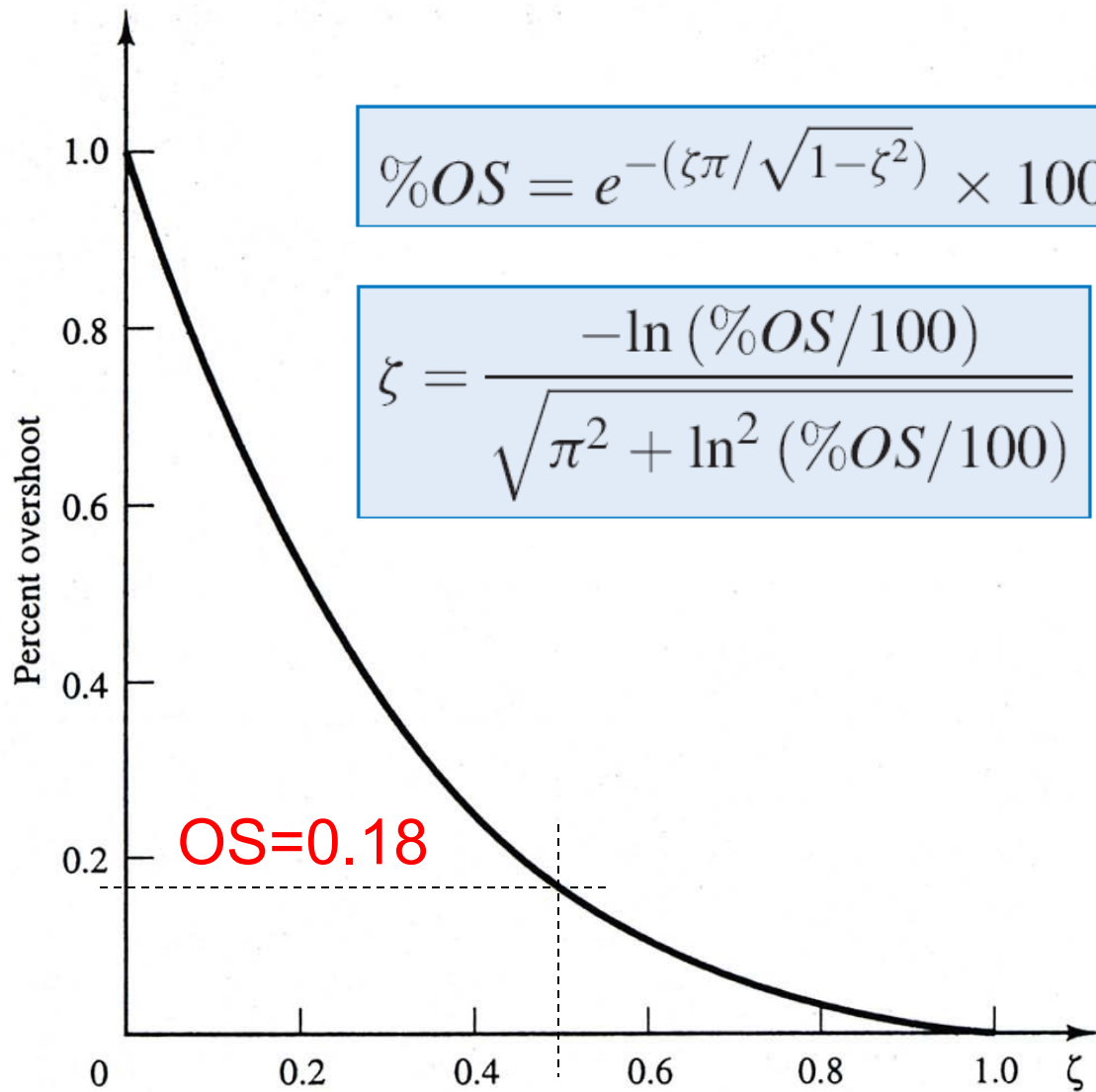


Figure 6-4 Percent overshoot versus ζ .

$\zeta=0.5$

•Approach-2: Diff. Eq.

The system time response can also be calculated using a difference-equations approach. From the expression for the closed-loop transfer function

$$\frac{C(z)}{R(z)} = \frac{0.368z^{-1} + 0.264z^{-2}}{1 - z^{-1} + 0.632z^{-2}} \quad (6-2)$$

or

$$C(z)[1 - z^{-1} + 0.632z^{-2}] = R(z)[0.368z^{-1} + 0.264z^{-2}] \quad (6-3)$$

Taking the inverse z-transform of (6-3), we obtain the difference equation

$$\begin{aligned} \text{Z}^{-1}: \text{Use Others } c(kT) &= 0.368r[(k-1)T] + 0.264r[(k-2)T] \\ &\quad + c[(k-1)T] - 0.632c[(k-2)T] \end{aligned} \quad (6-4)$$

Both $c(kT)$ and $r(kT)$ are zero for $k < 0$. Thus, from (6-4), $c(0) = 0$ and $c(1) = 0.368$.

For $k \geq 2$, (6-4) becomes

$$\underline{c(kT)} = 0.632 + c[(k-1)T] - 0.632c[(k-2)T]$$

6.3 System Characteristic Equation

Consider a single-loop sampled-data system of the type shown in Figure 6-5. For this system,

$$C(z) = \frac{G(z)R(z)}{1 + \overline{GH}(z)} = \frac{K \prod^m (z - z_i)}{\prod^n (z - p_i)} R(z)$$

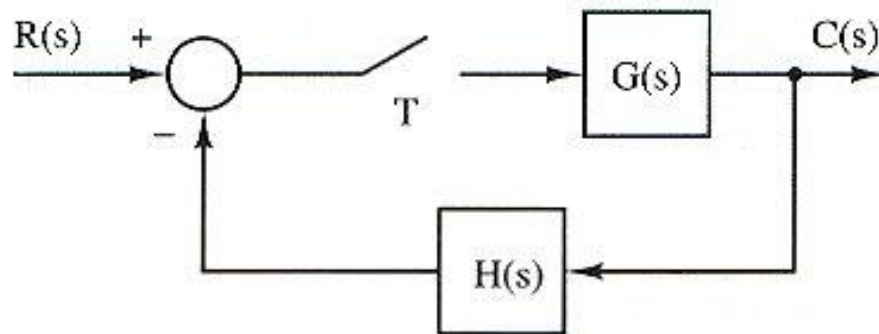


Figure 6-5 Single-loop sampled-data systems.

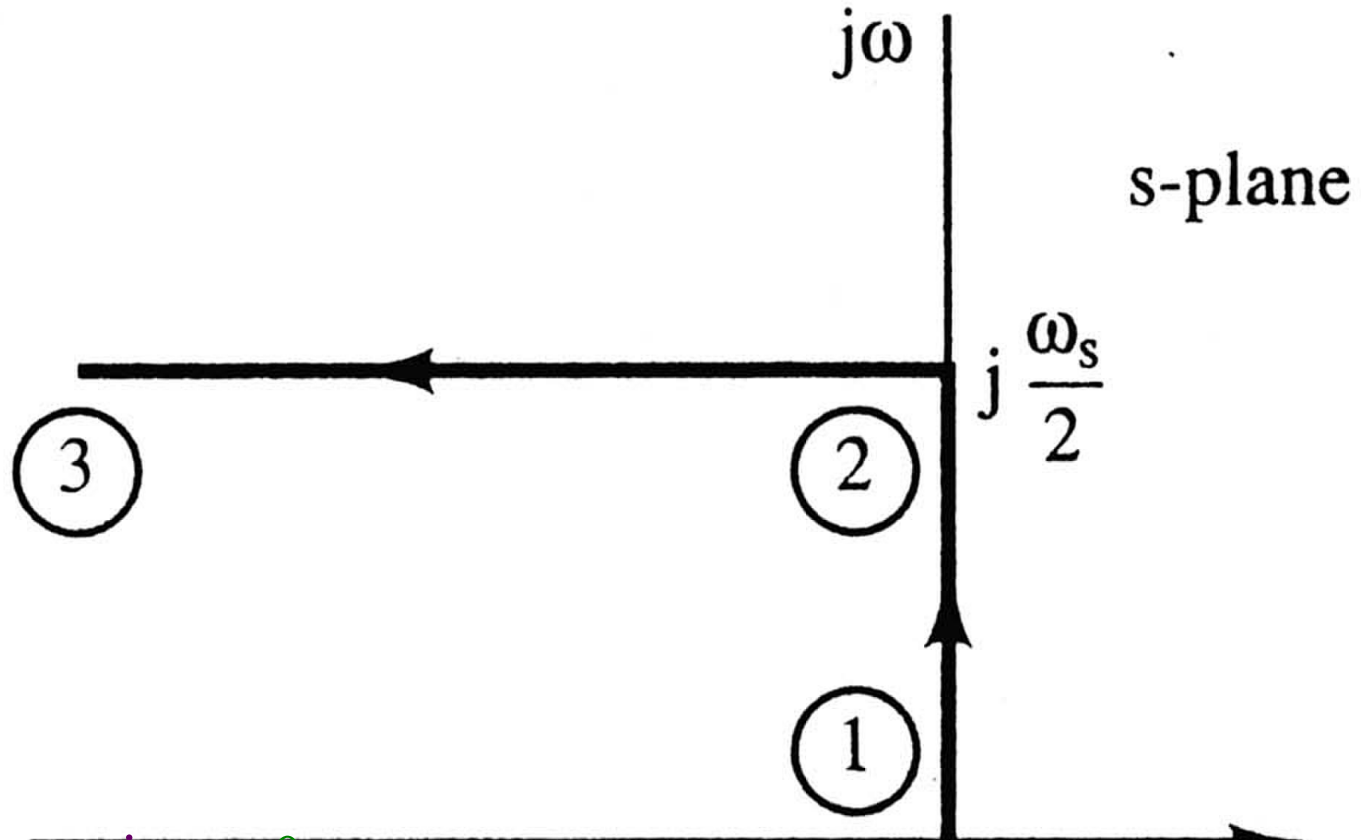
This equation is the system characteristic equation

$$1 + \overline{GH}(z) = 0$$

6.4 Mapping The s-Plane Into The z-Plane

- $E(s)$, $E^*(s)$, $E(z)$

$$E(z) = E^*(s) \big|_{e^{sT} = z}$$



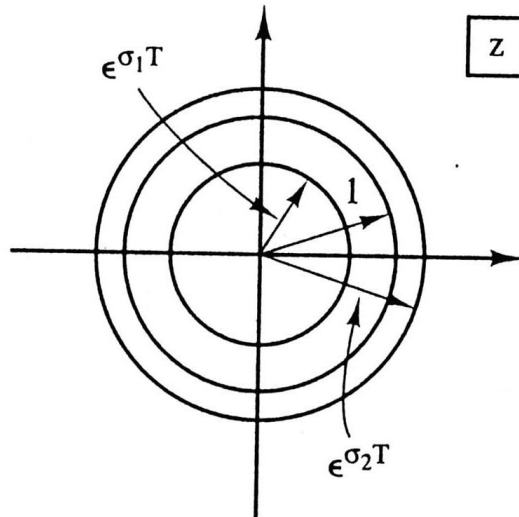
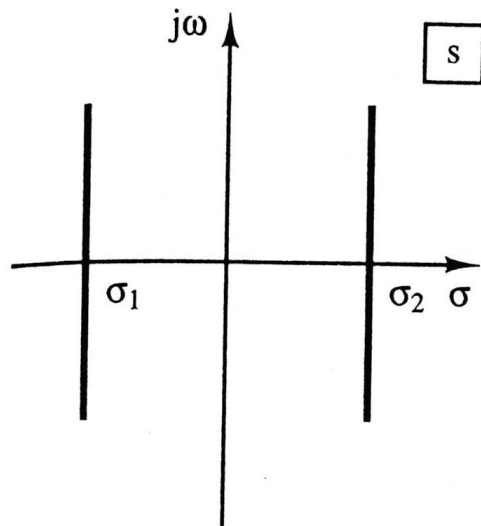
Along the $j\omega$ axis: $\sigma=0$

On the unit circle: $|z|=1$

$$z = e^{sT} = e^{\sigma T} e^{j\omega T} = \underline{e^{j\omega T}} = \cos \omega T + j \sin \omega T = \underline{1 \angle \omega T}$$

1) Constant Damping Loci

σ_d : Exponential Damping Frequency



Circles
($j\omega$ axis: $|z|=1$)

Figure 6-7 Mapping constant damping loci into the z-plane.

$$z = e^{\sigma_1 T} \underline{e^{j\omega T}} = e^{\sigma_1 T} \underline{\angle \omega T}$$

2) Constant Frequency Loci

ω_d : Damped Frequency of Oscillation

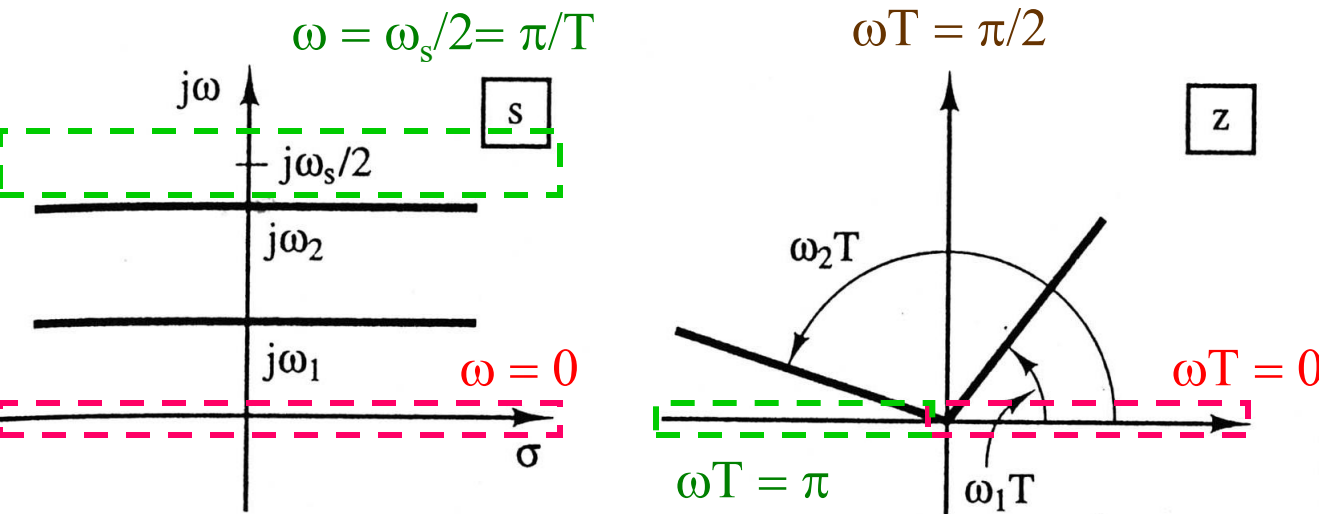
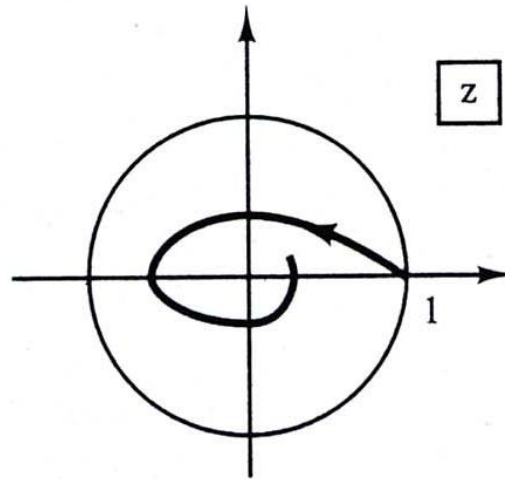
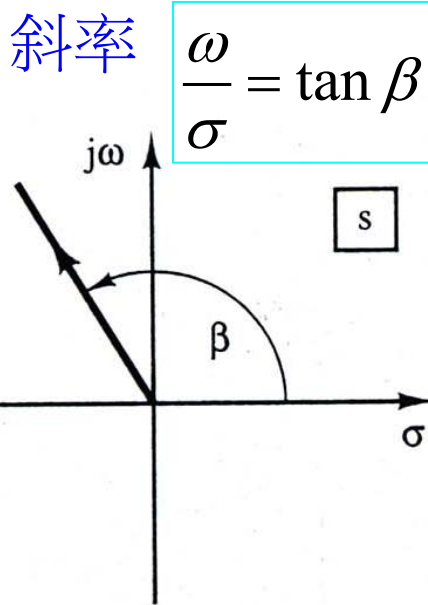


Figure 6-8 Mapping constant frequency loci into the z-plane.

Lines
(with angle= ωT)

3) Constant Damping Ratio Loci: ζ



σ 在 II-III quadrants 為負值，
所以 $e^{\sigma T}$ 漸小。

Figure 6-9 Mapping constant damping-ratio loci into the z-plane.

Constant damping ratio

$$\frac{\omega}{\sigma} = \tan \beta$$

$$\omega = \underline{\sigma \tan \beta}$$

$$z = e^{sT} = e^{\sigma T} \angle \underline{\sigma T \tan \beta} \quad (6.6)$$

Supplementary Materials

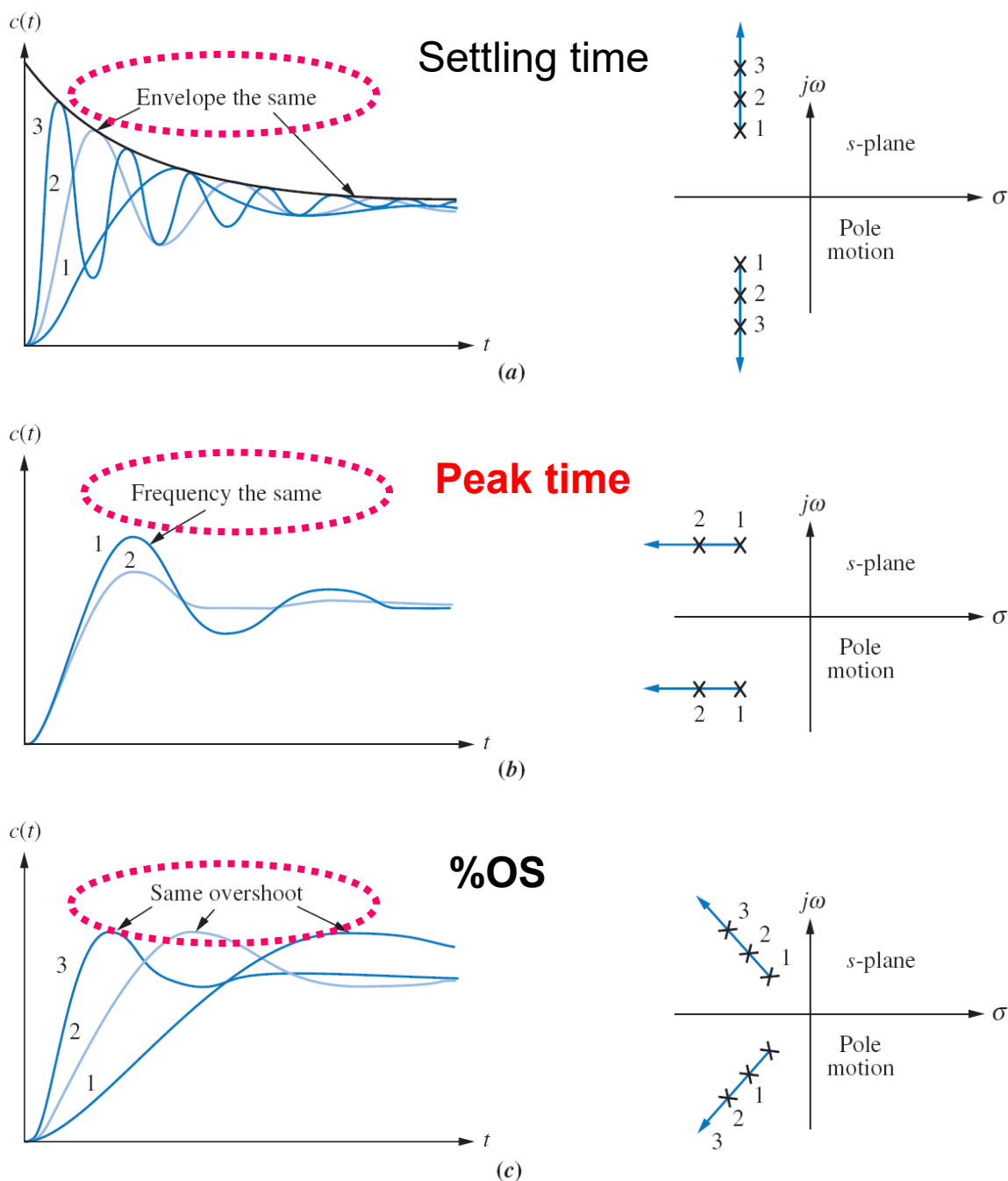
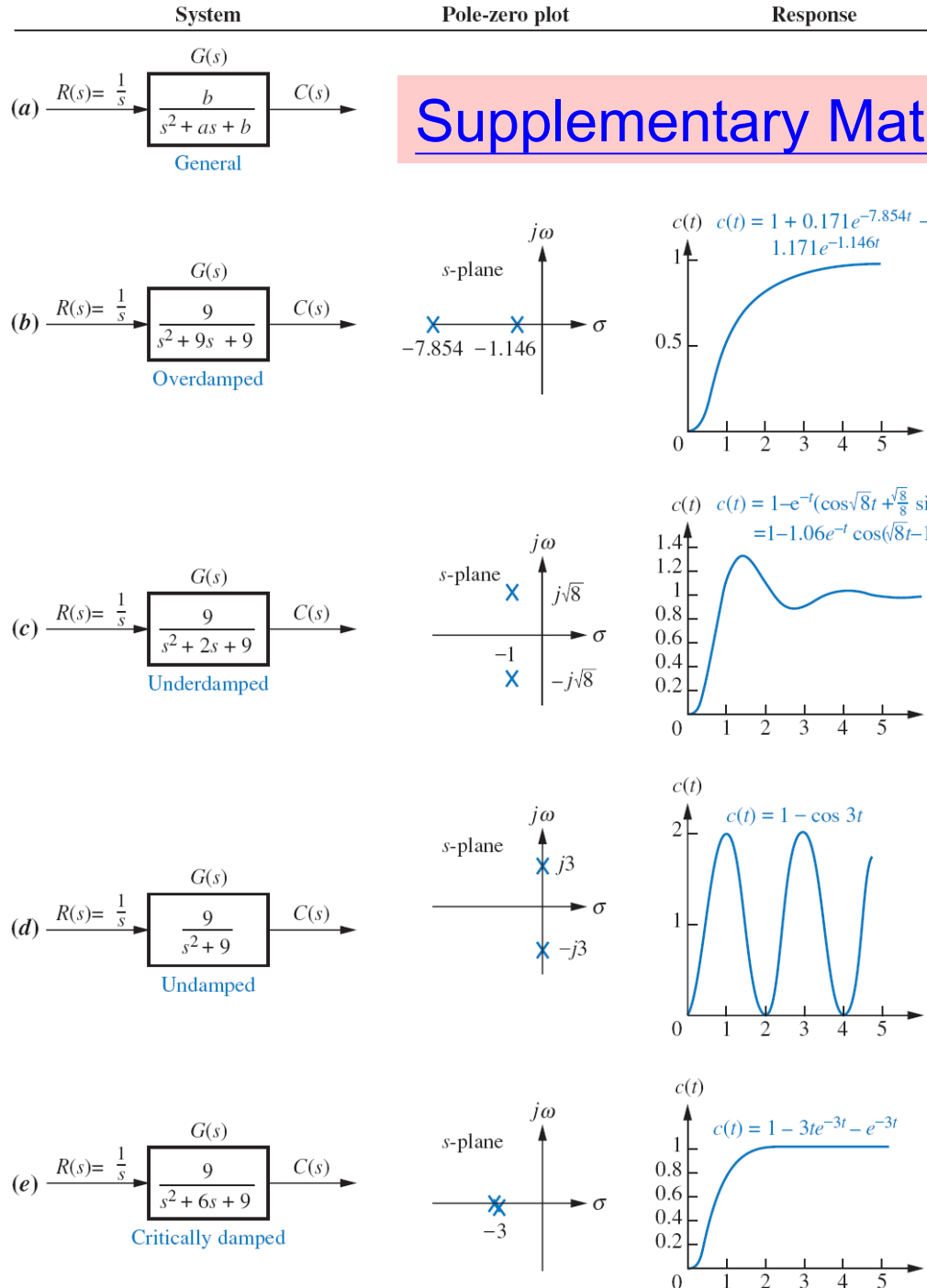


FIGURE 4.7 Second-order systems, pole plots, and step responses

Supplementary Materials



Real & distinct roots
(P35)

Complex or
Imaginary roots
(P39)

Real & repeated roots
(P38)

Fastest resp. w/o OS.

2nd-ORDER TRANSFER FUNCTIONS VIA TESTING

1) %OS $\rightarrow \zeta$

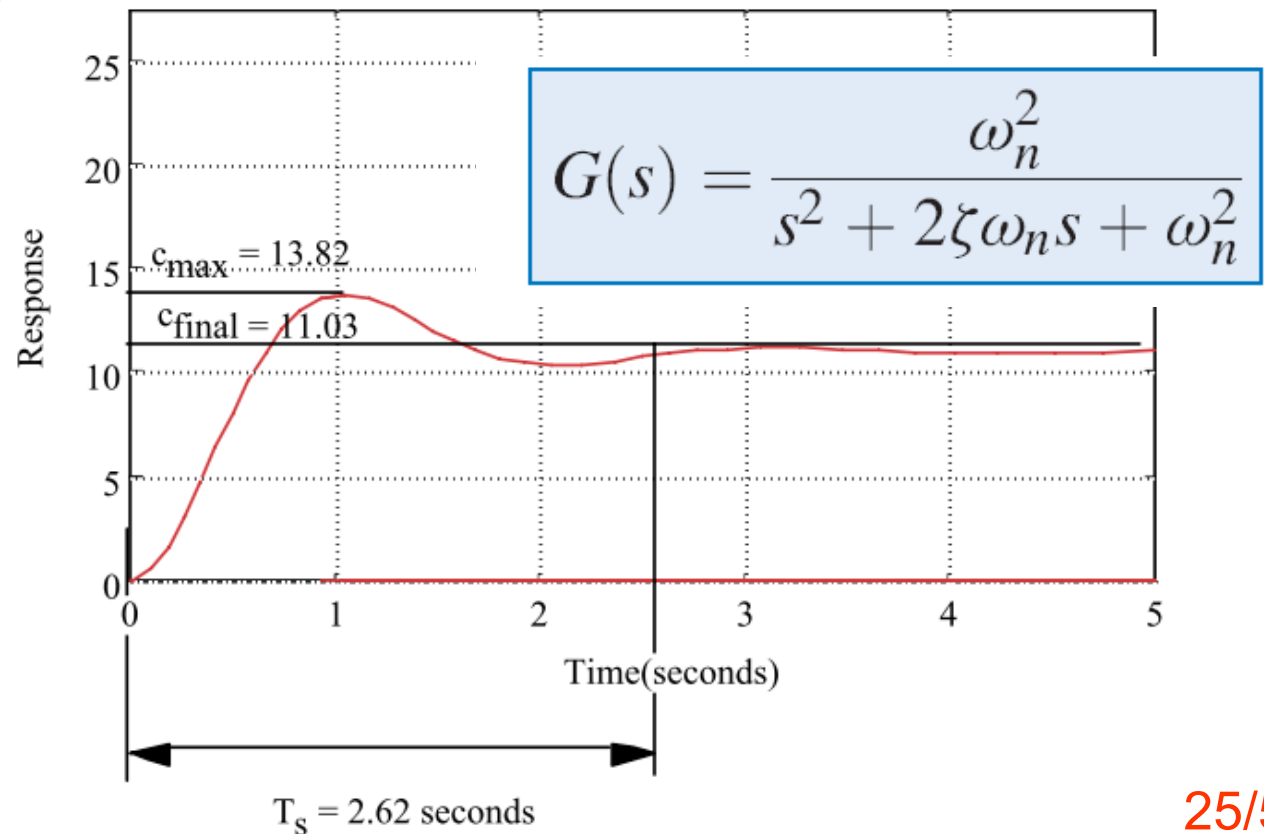
2) T_p or T_s (both with ζ) $\rightarrow \omega_n$

3) $c_{final} = K / \omega_n^2$

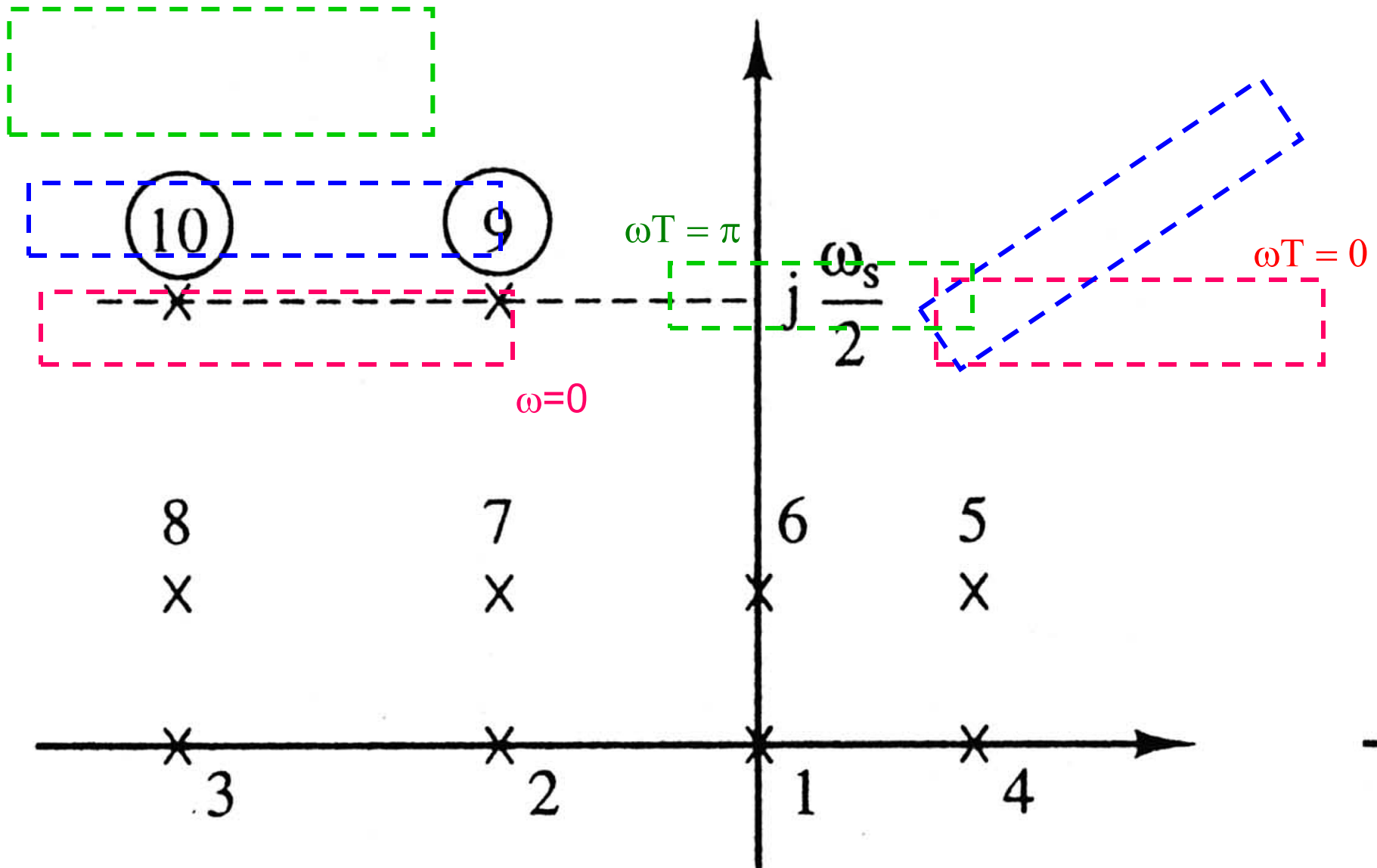
$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$T_s = \frac{4}{\zeta \omega_n}$$



$$\omega = \omega_s/2 = \pi/T$$



If (2nd-Order Syst.)-damped

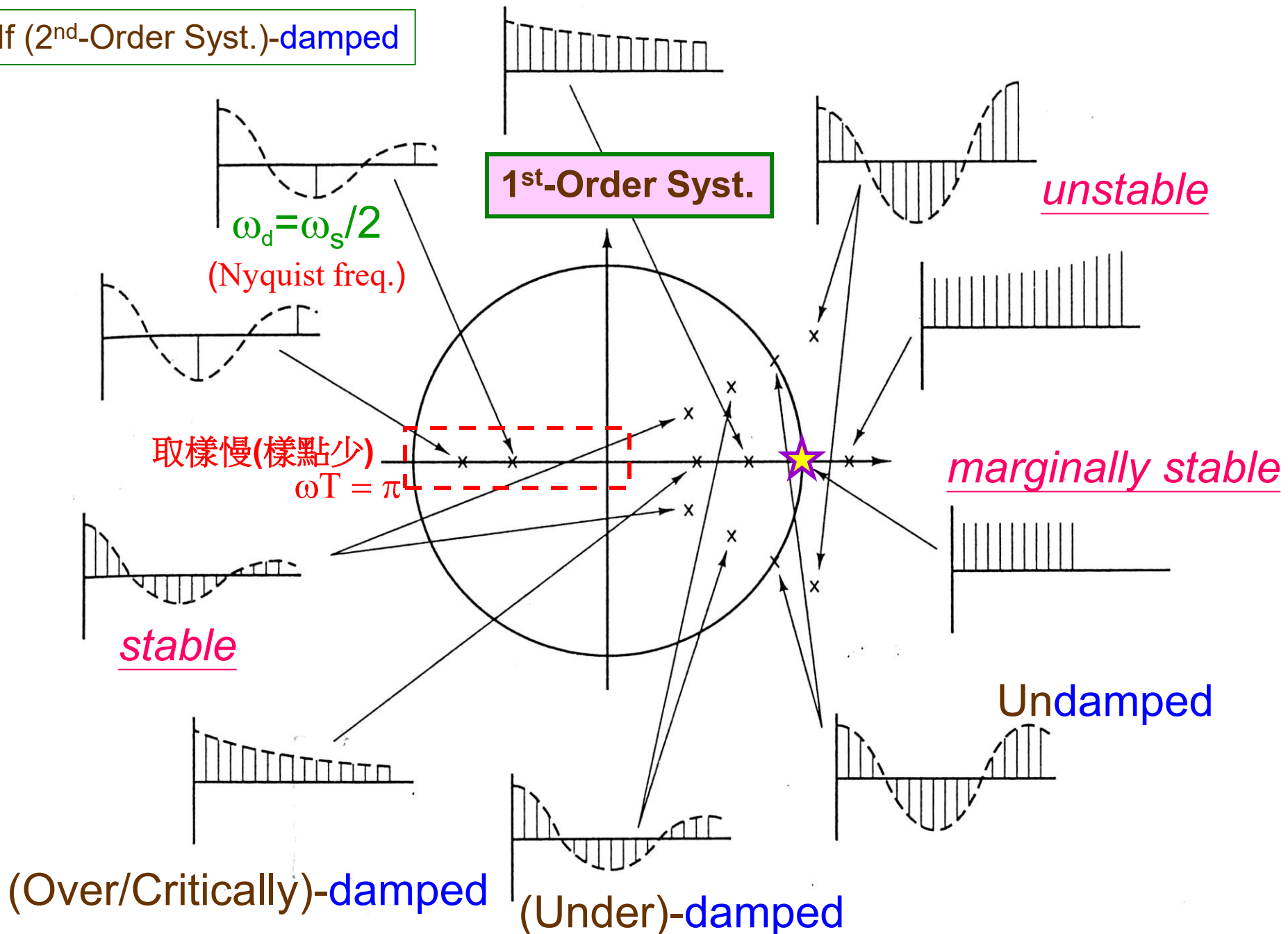


Figure 6-11 Transient response characteristics of the z-plane pole locations.

Given z-plane pole location,
find ζ , ω_n , and τ (T_c)

Standard form the S-plane 2nd-order
transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Which has the poles $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

The equivalent Z-plane
poles occur at

$$z = e^{sT} \Big|_{s_{1,2}} = e^{-\zeta\omega_n T} \angle \pm \omega_n T \sqrt{1-\zeta^2} = r \angle \pm \theta$$

Hence $e^{-\zeta\omega_n T} = r$ or $\zeta\omega_n T = -\ln r$

Also $\omega_n T \sqrt{1-\zeta^2} = \theta$

Taking the ratio of the last two equations $\frac{\zeta}{\sqrt{1-\zeta^2}} = \frac{-\ln r}{\theta} \Rightarrow \zeta = \frac{-\ln r}{\sqrt{\ln^2 r + \theta^2}} \quad (6.8)$

$$\omega_n = \frac{1}{T} \sqrt{\ln^2 r + \theta^2} \quad (6.9)$$

Time constant $\tau = \frac{1}{\zeta\omega_n} = \frac{-T}{\ln r} \quad (6.10)$

Expressed as $r = e^{-T/\tau}$

Example 6.6

For this example we will consider the system of [Example 6.4](#). For this system the closed-loop transfer function was calculated to be

$$\frac{G(z)}{1 + G(z)} = \frac{0.368z + 0.264}{z^2 - z + 0.632}, \quad T = 1 \text{ s}$$

Thus the system characteristic equation is

$$z^2 - z + 0.632 = (z - 0.5 - j0.618)(z - 0.5 + j0.618) = 0$$

The poles are then complex and occur at

$$z = 0.5 \pm j0.618 = 0.795 \angle \pm 51.0^\circ = 0.795 \angle \pm 0.890 \text{ rad}$$

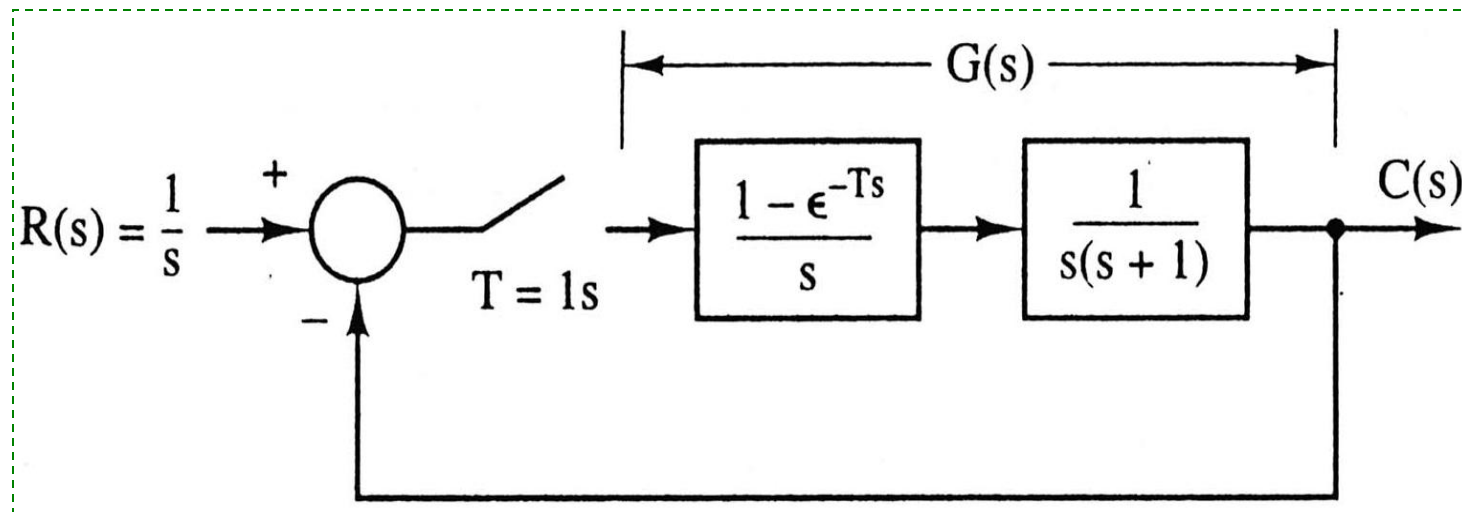


Fig. 6-2

Since, in (6-8), (6-9), and (6-10),

$$z = e^{sT} / \pm \omega T = r / \pm \omega T = 0.795 / \pm 0.890$$

then

$$\zeta = \frac{-\ln(0.795)}{[\ln^2(0.795) + (0.890)^2]^{1/2}} = 0.250$$

$$\omega_n = \frac{1}{1} [\ln^2(0.795) + (0.890)^2]^{1/2} = 0.9191$$

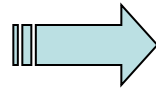
$$\tau = \frac{-1}{\ln(0.795)} = 4.36 \text{ s}$$

CT viewpoint

The equivalent values for the closed-loop analog system (with sampler and data hold removed) were found in Example 6.4 to be $\zeta = 0.50$ and $\omega_n = 1 \text{ rad/s}$. Also $\tau = 1/\zeta\omega_n = 2 \text{ s}$. Hence the effects of the sampling are seen to be destabilizing. However, if $T = 0.1 \text{ s}$, there is little effect from the sampling. From Problem 6.10, the parameters for this case are $\zeta = 0.475$, $\omega_n = 0.998 \text{ rad/s}$, and $\tau = 2.11 \text{ s}$.

To show the effect of sampling: destabilizing effects!

$$\tau = \frac{1}{\zeta \omega_n} = \frac{-T}{\ln r} \quad (6.10)$$



$$\frac{\tau}{T} = -\frac{1}{\ln r} \quad (6.11)$$

The number of samples per time constant
Ex. If $\tau=1$, $T=0.25$, then $1/0.25 = 4$

Summary:

For real poles

$$T \ll \tau$$

For complex poles, more
+ $\omega T < 1$ rad. (~57 deg.)

Put poles to the vicinity of $z = 1$

TABLE 6-2

<u>Samples per time constant</u>		<u>Samples per period</u>	
r	τ/T	θ	T_d/T
0.999	999.5	10°	36
0.99	99.5	20°	18
0.95	19.5	30°	12
0.9	9.5	45°	8
0.8	4.48	60°	6
0.7	2.80	90°	4
0.6	1.96	120°	3
0.4	1.09	180°	2
0.2	0.62		

T_d : the period of the sinusoid

$$\frac{T_d}{T} = \frac{2\pi}{\theta} = \frac{360^\circ}{\theta^\circ}$$

(6.12)

$$\omega_d = \omega_s/2$$

See slide: P27. 1/56



Shannon's Sampling Theorem (1949)

- If a function $e(t)$ contains **no frequencies higher than f_0 hertz**, it is completely determined by giving its ordinates at a series of points spaced **$1/(2f_0)$ seconds** apart.
 - $e(t)$ is **band-limited** to one-sided baseband bandwidth f_0
 - f_s : **sampling rate**

$$f_s > 2f_0$$

Nyquist rate: $2f_0$
Nyquist freq: $f_s/2$

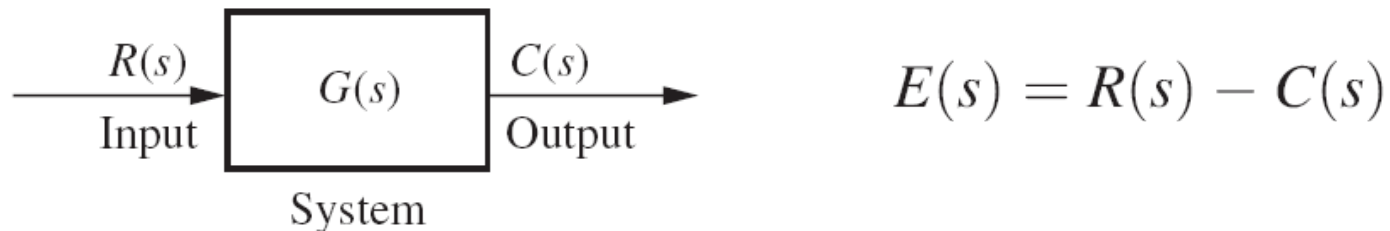
- The theorem is commonly called the **Nyquist sampling theorem**.
- But, why Nyquist?

6.5 Steady-State Accuracy

START: Supplementary Materials from Nise Book!

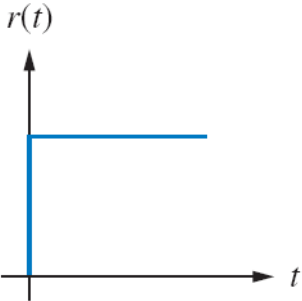
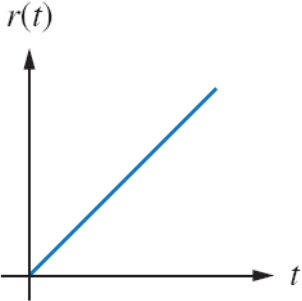
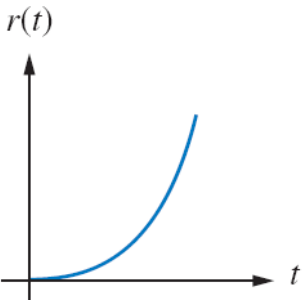
7.1 INTRODUCTION

- DEFINITION AND TEST INPUTS
 - *Steady-state error*: the difference between the input and the output for a prescribed test input as $t \rightarrow \infty$.



- APPLICATION TO STABLE SYSTEMS

TABLE 7.1 Test waveforms for evaluating steady-state errors of position control systems

Waveform	Name	Physical interpretation	Time function	Laplace transform
 A graph of a step function $r(t)$ versus time t. The vertical axis is labeled $r(t)$ and the horizontal axis is labeled t. The function is zero for $t < 0$ and jumps to a constant positive value for $t \geq 0$.	Step	Constant position	1	$\frac{1}{s}$
 A graph of a ramp function $r(t)$ versus time t. The vertical axis is labeled $r(t)$ and the horizontal axis is labeled t. The function is zero for $t < 0$ and increases linearly with a constant slope for $t \geq 0$.	Ramp	Constant velocity	t	$\frac{1}{s^2}$
 A graph of a parabolic function $r(t)$ versus time t. The vertical axis is labeled $r(t)$ and the horizontal axis is labeled t. The function is zero for $t < 0$ and increases quadratically for $t \geq 0$.	Parabola	Constant acceleration	$\frac{1}{2}t^2$	$\frac{1}{s^3}$

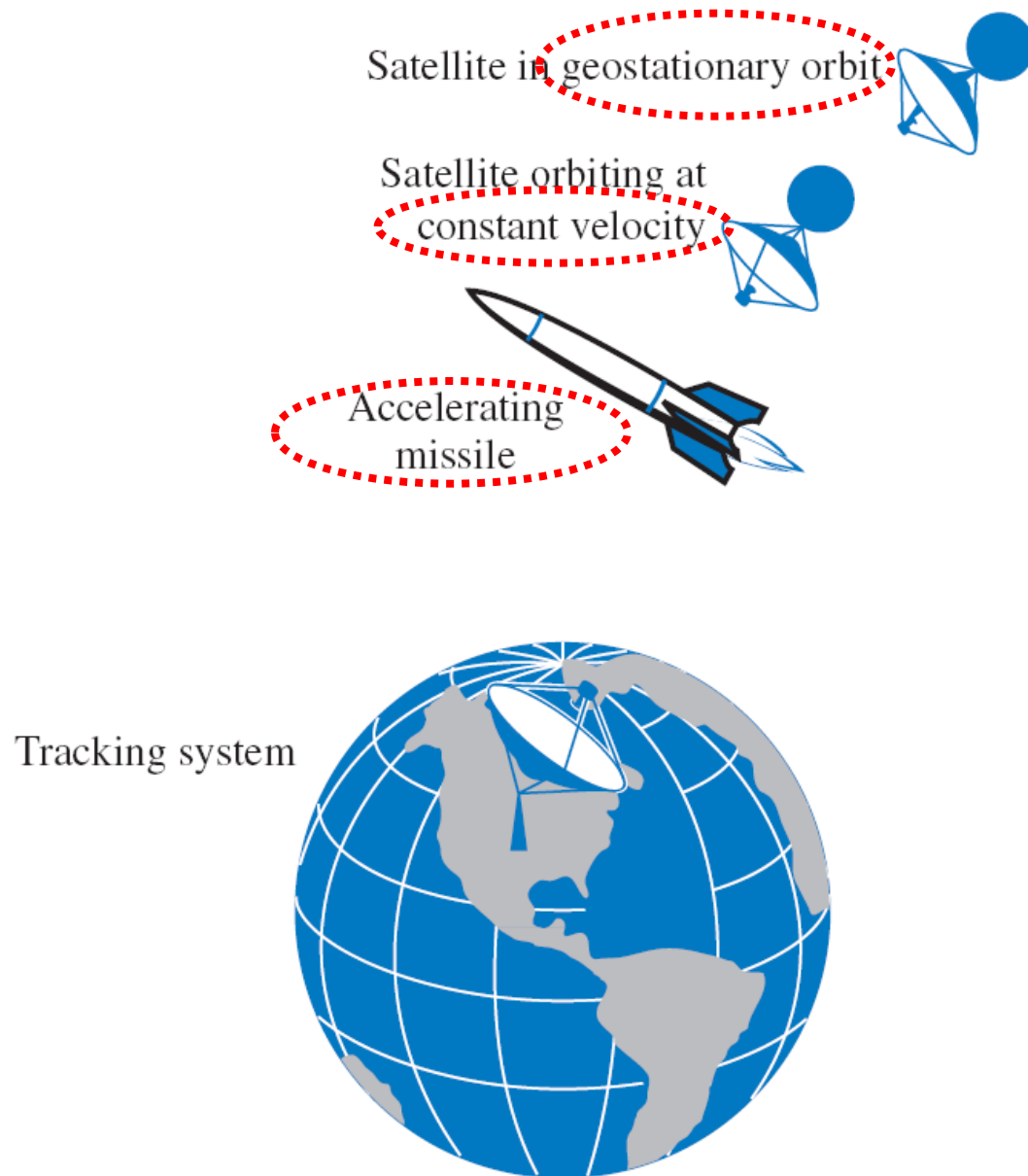
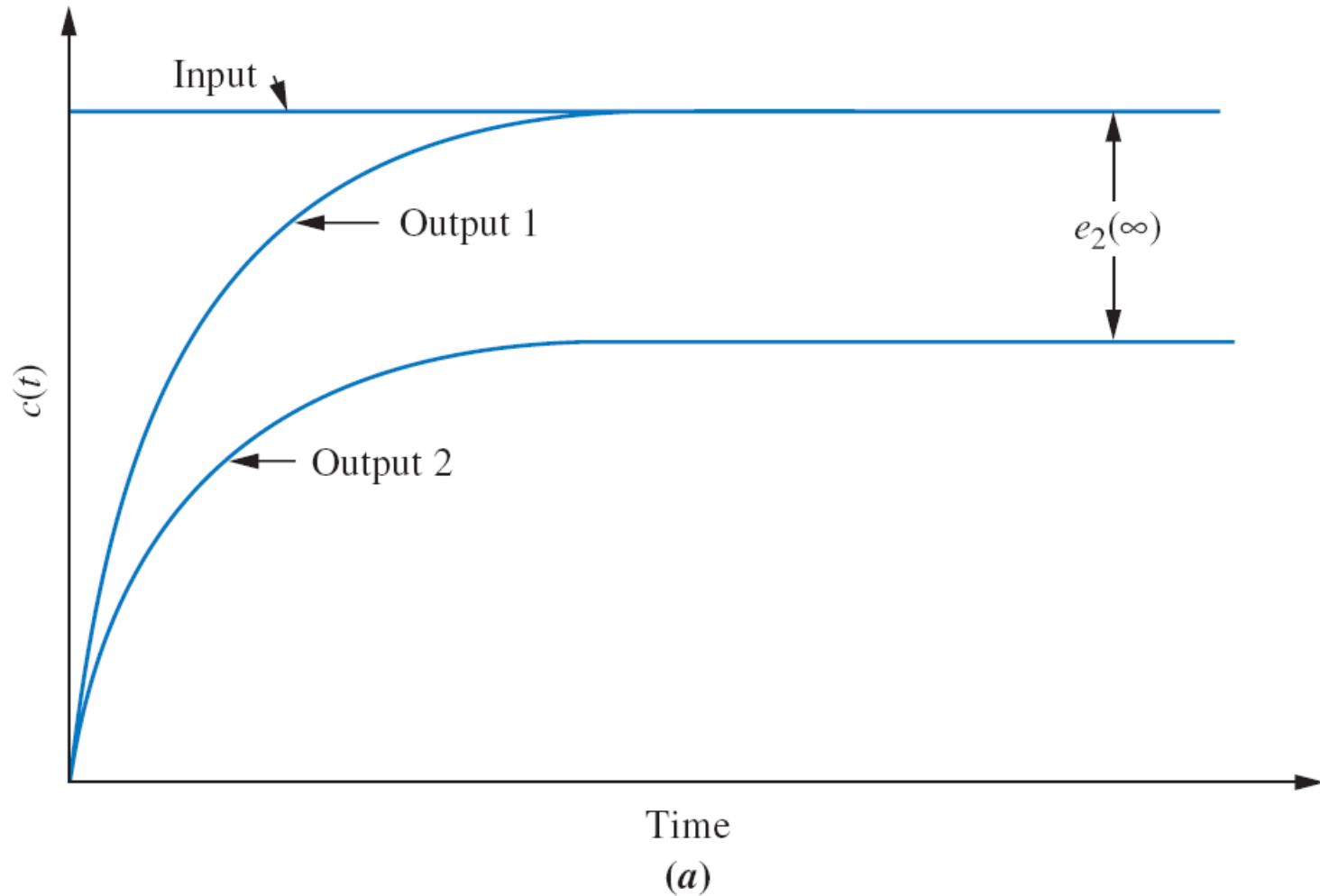


FIGURE 7.1 Test inputs for steady-state error analysis and design vary with target type

EVALUATING STEADY-STATE ERRORS



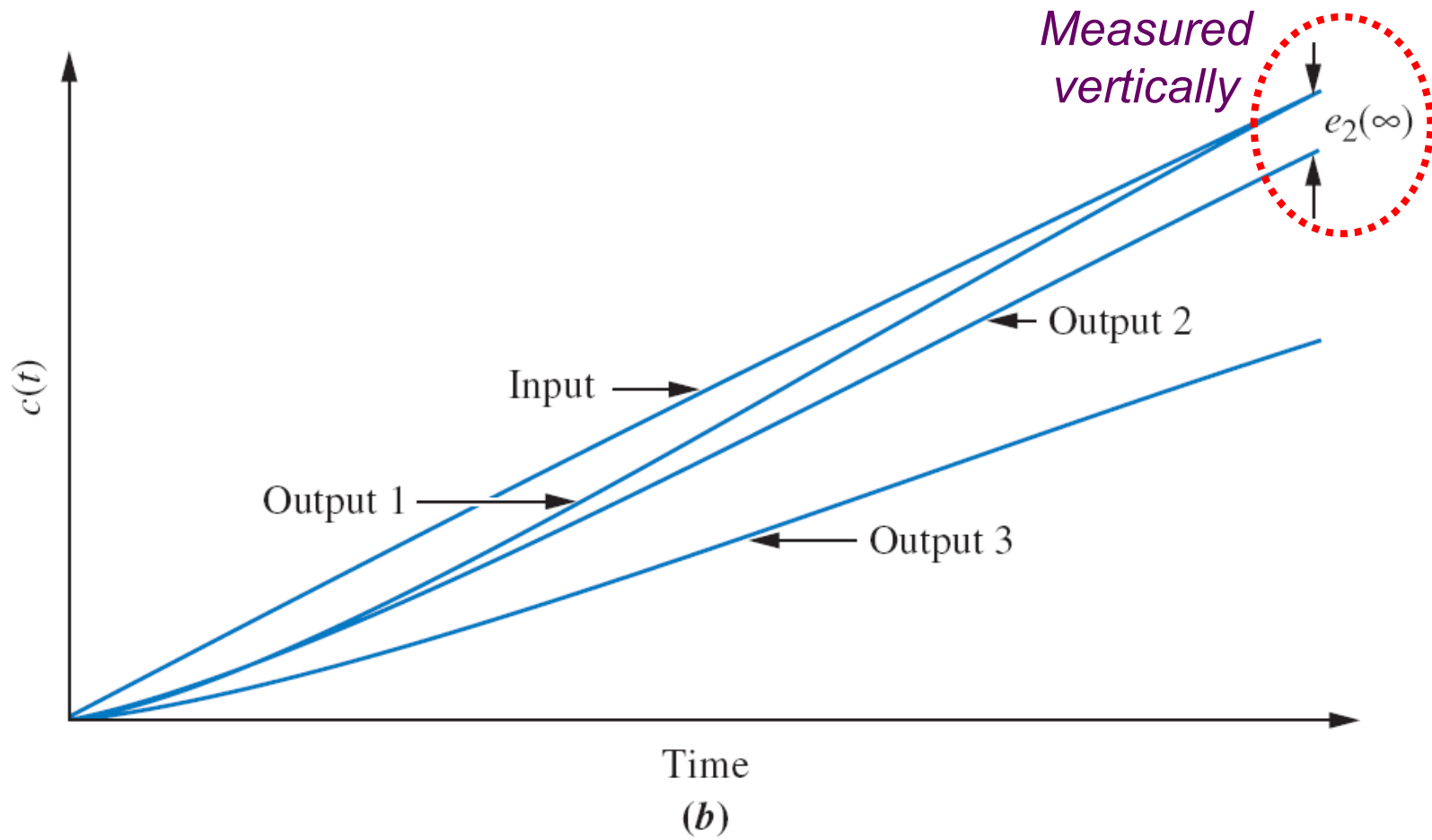


FIGURE 7.2 Steady-state error: **a.** step input; **b.** ramp input

SOURCES OF STEADY-STATE ERROR

- Error Sources
 1. Nonlinearity
 2. System Configurations
 3. Disturbance...

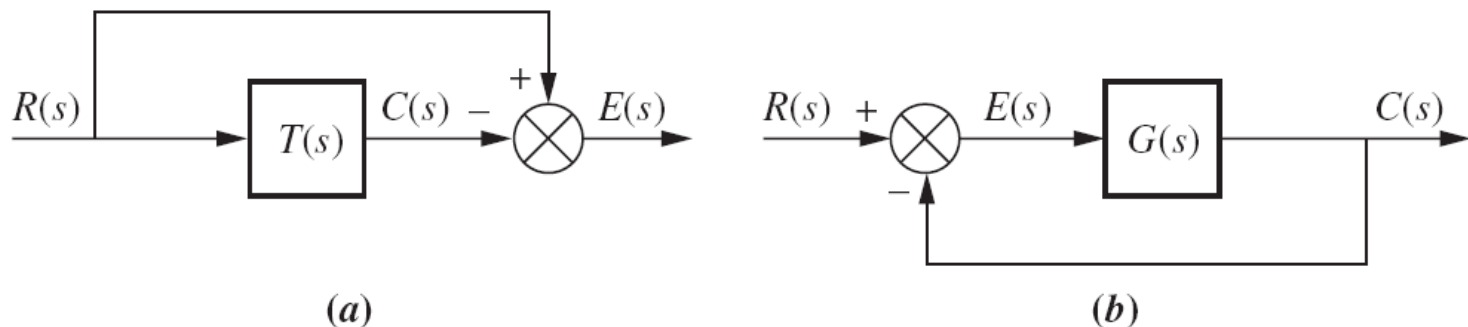


FIGURE 7.3 Closed-loop control system error: **a.** general representation; **b.** representation for unity feedback systems

System Configurations

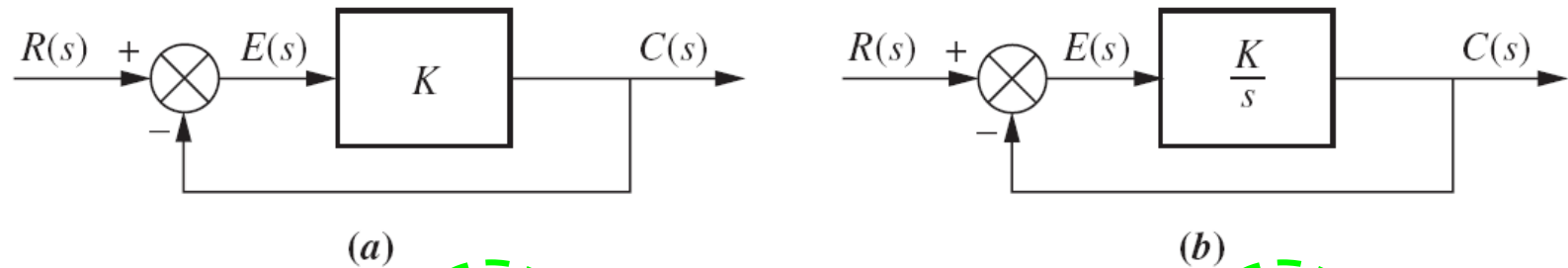


FIGURE 7.4 System with **a** (finite) steady-state error for a step input; **b** (zero) steady-state error for step input

If $c(t)$ is to be finite and nonzero,

$$\text{a) } e_{\text{steady-state}} = \frac{1}{K} c_{\text{steady-state}} \quad (7.1)$$

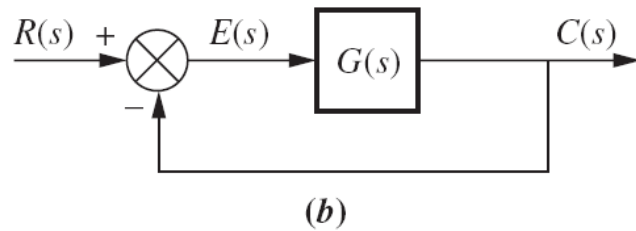
$$\text{b) } E(s) = \frac{1}{K} sC(s)$$

$$\Rightarrow e(\infty) = \frac{1}{K} c'(\infty) = 0$$

Unity feedback

- STEADY-STATE ERROR IN TERMS OF $G(s)$

$$E(s) = R(s) - C(s) \quad (7.8)$$



$$C(s) = E(s)G(s) \quad (7.9)$$

$$E(s) = \frac{R(s)}{1 + G(s)} \quad (7.10)$$

For **stable** systems:



$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)} \quad (7.11)$$

Step input

$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)} \quad (7.12)$$

dc gain

Zero error



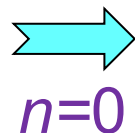
$$\lim_{s \rightarrow 0} G(s) = \infty \quad (7.13)$$

(see Tab 2.2, Item 10)

$$G(s) \equiv \frac{(s + z_1)(s + z_2) \cdots}{s^n (s + p_1)(s + p_2) \cdots} \quad (7.14)$$

$n \geq 1$, i.e. at least one **pure integration** in the forward path.

Finite error
(no integration)



$n=0$

$$\lim_{s \rightarrow 0} G(s) = \frac{z_1 z_2 \cdots}{p_1 p_2 \cdots} \quad (7.15)$$

Ramp Input

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)} \quad (7.16)$$

Zero error  $\lim_{s \rightarrow 0} sG(s) = \infty$ (7.17)

$n \geq 2$, i.e. at least two pure integrations in form of (7.14)

Finite error
(1 integration)  $\lim_{s \rightarrow 0} sG(s) = \frac{z_1 z_2 \cdots}{p_1 p_2 \cdots}$ (7.18)
 $n=1$

Infinite error
(no integration)  $\lim_{s \rightarrow 0} sG(s) = 0$ (7.19)
 $n=0$

Parabolic Input

$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)} \quad (7.20)$$

Zero error $\Rightarrow \lim_{s \rightarrow 0} s^2 G(s) = \infty \quad (7.21)$

$n \geq 3$, i.e. at least three pure integrations in form of (7.14)

Finite error
(2 integrations) $\Rightarrow \lim_{s \rightarrow 0} s^2 G(s) = \frac{z_1 z_2 \cdots}{p_1 p_2 \cdots} \quad (7.22)$
 $n=2$

Infinite error
(1 or less) $\Rightarrow \lim_{s \rightarrow 0} s^2 G(s) = 0 \quad (7.23)$
 $n \leq 1$

7.3 STATIC ERROR CONSTANTS AND SYSTEM TYPE

- STATIC ERROR CONSTANTS

Test Inputs

$u(t)$

$$e(\infty) = e_{\text{step}}(\infty) = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)} \quad (7.30)$$

$tu(t)$

$$e(\infty) = e_{\text{ramp}}(\infty) = \frac{1}{\lim_{s \rightarrow 0} sG(s)} \quad (7.31)$$

$1/2 t^2 u(t)$

$$e(\infty) = e_{\text{parabola}}(\infty) = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)} \quad (7.32)$$

STATIC ERROR CONSTANTS

position constant, K_p , where

$$K_p = \lim_{s \rightarrow 0} G(s) \quad (7.33)$$

velocity constant, K_v , where

$$K_v = \lim_{s \rightarrow 0} sG(s) \quad (7.34)$$

acceleration constant, K_a , where

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) \quad (7.35)$$

As $K_{p,v,a} \uparrow$
 $\rightarrow e(\infty) \downarrow$.

SYSTEM TYPE

Type n

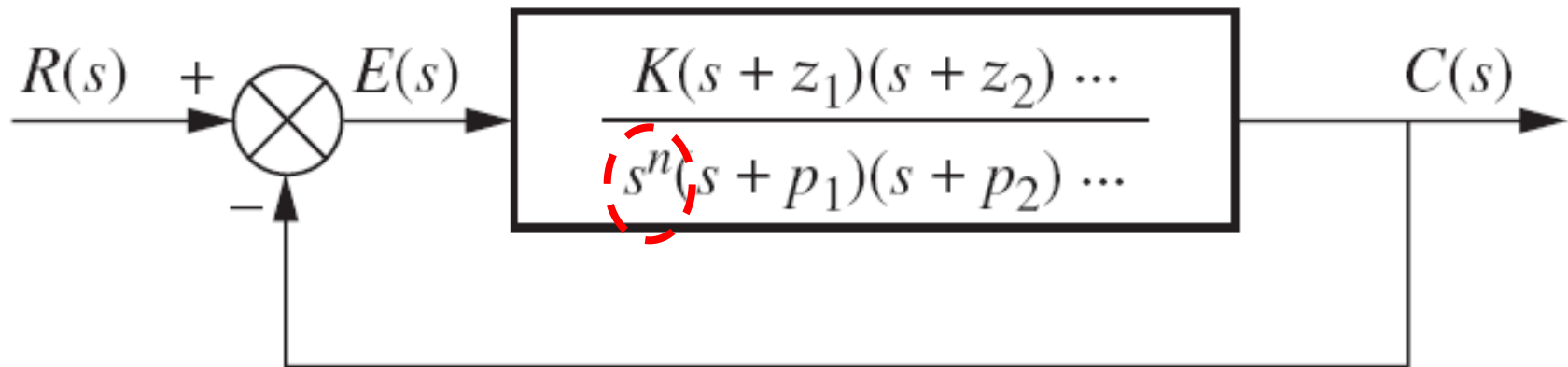


FIGURE 7.8 Feedback control system for defining system type

TABLE 7.2 Relationships between input, system type, static error constants, and steady-state errors

Input	Steady-state error formula	Type 0		Type 1		Type 2	
		Static error constant	Error	Static error constant	Error	Static error constant	Error
Step, $u(t)$	$\frac{1}{1 + K_p}$	$K_p = \text{Constant}$	$\frac{1}{1 + K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = \text{Constant}$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = \text{Constant}$	$\frac{1}{K_a}$

7.4 STEADY-STATE ERROR SPECIFICATIONS

- ❖ Example: Spec. $K_v=1000$ means
 1. Stable system
 2. Type 1
 3. Ramp input (test signal)
 4. Steady-state error is $1/K_v$.

END: Supplementary Materials from Nise Book!

6.5 Steady-State Accuracy

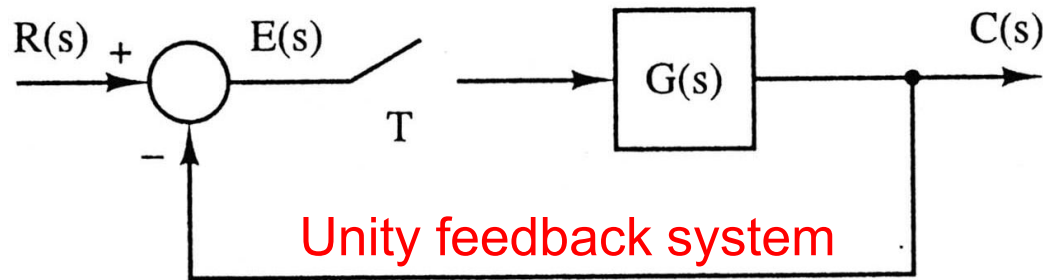


Figure 6-12 Discrete-time system.

$$\frac{C(z)}{R(z)} = \frac{G(z)}{1 + G(z)} \quad (6.13)$$

The plant transfer function can always be expressed as

$$G(z) = \frac{K \prod_{i=1}^m (z - z_i)}{(z - 1)^N \prod_{j=1}^P (z - z_j)}, z_i \neq 1, z_j \neq 1 \quad (6.14)$$

Define $K_{dc} = \left. \frac{K \prod_{i=1}^m (z - z_i)}{\prod_{j=1}^P (z - z_j)} \right|_{z=1}$

Open-loop plant dc gain with all poles at $z=1$ removed.

$$(6.15)$$

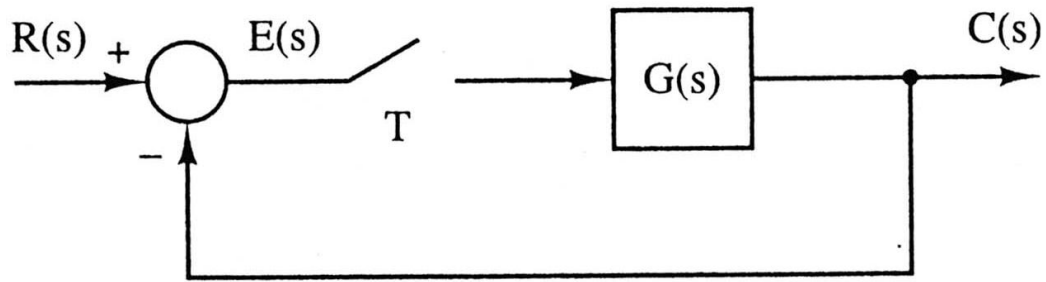


Figure 6-12 Discrete-time system.

$$E(z) = R(z) - C(z) \quad (6.16)$$

$$E(z) = R(z) - \frac{G(z)}{1 + G(z)} R(z) = \frac{R(z)}{1 + G(z)}$$

Steady-state error $e_{ss}(kT) = \lim_{z \rightarrow 1} (z - 1)E(z) = \lim_{z \rightarrow 1} \frac{(z - 1)R(z)}{1 + G(z)} \quad (6.17)$

For the step-input $R(z) = \frac{z}{z - 1}$

$$e_{ss}(kT) = \lim_{z \rightarrow 1} \frac{z}{1 + G(z)} = \frac{1}{1 + \lim_{z \rightarrow 1} G(z)}$$

Define the **position error constant, K_p**

$$K_p = \lim_{z \rightarrow 1} G(z) \quad (6.18) \quad \Rightarrow \quad e_{ss}(kT) = \frac{1}{1 + K_p} = \frac{1}{1 + K_{dc}} \quad (6.19)$$

For $N \geq 1$, $K_p = \infty$, $e_{ss} = 0$.

For the unit-ramp input

$$R(z) = \frac{Tz}{(z-1)^2}$$

Steady-state error

$$e_{ss}(kT) = \lim_{z \rightarrow 1} \frac{Tz}{(z-1) + (z-1)G(z)} = \frac{T}{\lim_{z \rightarrow 1} (z-1)G(z)}$$

Define the **velocity error constant, K_v**

$$K_v = \lim_{z \rightarrow 1} \frac{1}{T} (z-1)G(z) \quad (6.20)$$

If $N=0$, $K_v=0$, $e_{ss}=\infty$.

$$\Rightarrow \quad e_{ss}(kT) = \frac{1}{K_v} = \frac{T}{K_{dc}} \quad (6.21)$$

For $N \geq 2$, $K_v = \infty$, $e_{ss} = 0$.

Example 6.7

Analysis

The steady-state errors will be calculated for the system of Figure 6-12, in which the open-loop function is given as

$$G(s) = \frac{1 - \epsilon^{-Ts}}{s} \left[\frac{K}{s(s+1)} \right]$$

$$G_p(s) = \frac{K}{s(s+1)}$$

Thus

$$\begin{aligned} G(z) &= K \mathfrak{Z} \left[\frac{1 - \epsilon^{-Ts}}{s^2(s+1)} \right] = \frac{K(z-1)}{z} \mathfrak{Z} \left[\frac{1}{s^2(s+1)} \right] \\ &= \frac{K(z-1)}{z} \frac{z[(\epsilon^{-T} + T - 1)z + (1 - \epsilon^{-T} - T\epsilon^{-T})]}{(z-1)^2(z - \epsilon^{-T})} \\ &= \frac{K[(\epsilon^{-T} + T - 1)z + (1 - \epsilon^{-T} - T\epsilon^{-T})]}{(z-1)(z - \epsilon^{-T})} \end{aligned}$$

Then, from (6-14) and (6-20), the system is type 1 and

$$K_v = \frac{K_{dc}}{T} = \frac{K[(\epsilon^{-T} + T - 1) + (1 - \epsilon^{-T} - T\epsilon^{-T})]}{T(1 - \epsilon^{-T})} = K$$

Since $G(z)$ has one pole at $z = 1$, the steady-state error to a step input is zero, and to a ramp input is, from (6-21),

$$e_{ss}(kT) = \frac{1}{K_v} = \frac{1}{K}$$

provided that the system is stable. The question of stability is considered in Chapter 7.



Example 6.8

Design

$K_v=100$

↑

As a second example, consider again the system of Figure 6-12, where, for this example,

$$G(z) = \mathcal{Z} \left[\frac{1 - e^{-Ts}}{s(s + 1)} \right] = \frac{1 - e^{-T}}{z - e^{-T}}$$

$$G_p(s) = \frac{1}{s + 1}$$

Suppose that the design specification for this system requires that the steady-state error to a unit ramp input be less than 0.01. Thus, from (6-20), it is necessary that the open-loop system be type 1 or greater and thus the open-loop function must have at least one pole at $z = 1$. Since the plant does not contain a pole at $z = 1$, a digital compensator of the form

$$D(z) = \frac{K_I z}{z - 1} + K_p$$

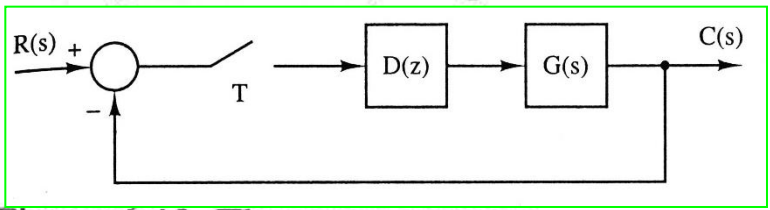


Fig. 6.13

will be added, to produce the resultant system shown in Figure 6-13. The compensator, called a PI or proportional-plus-integral compensator, is of a form commonly used to reduce steady-state errors. This compensator is discussed in Chapter 8. For this system (6-20) becomes

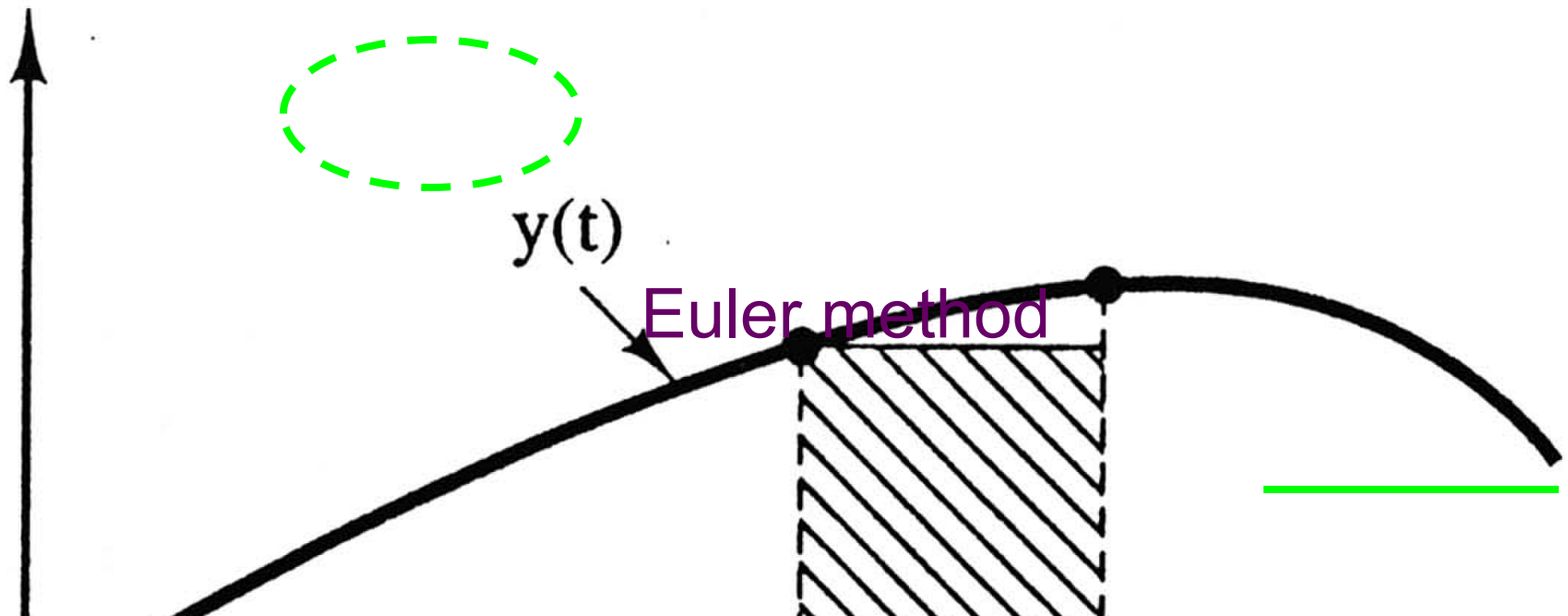
$$K_v = \lim_{z \rightarrow 1} \frac{1}{T} (z - 1) D(z) G(z)$$

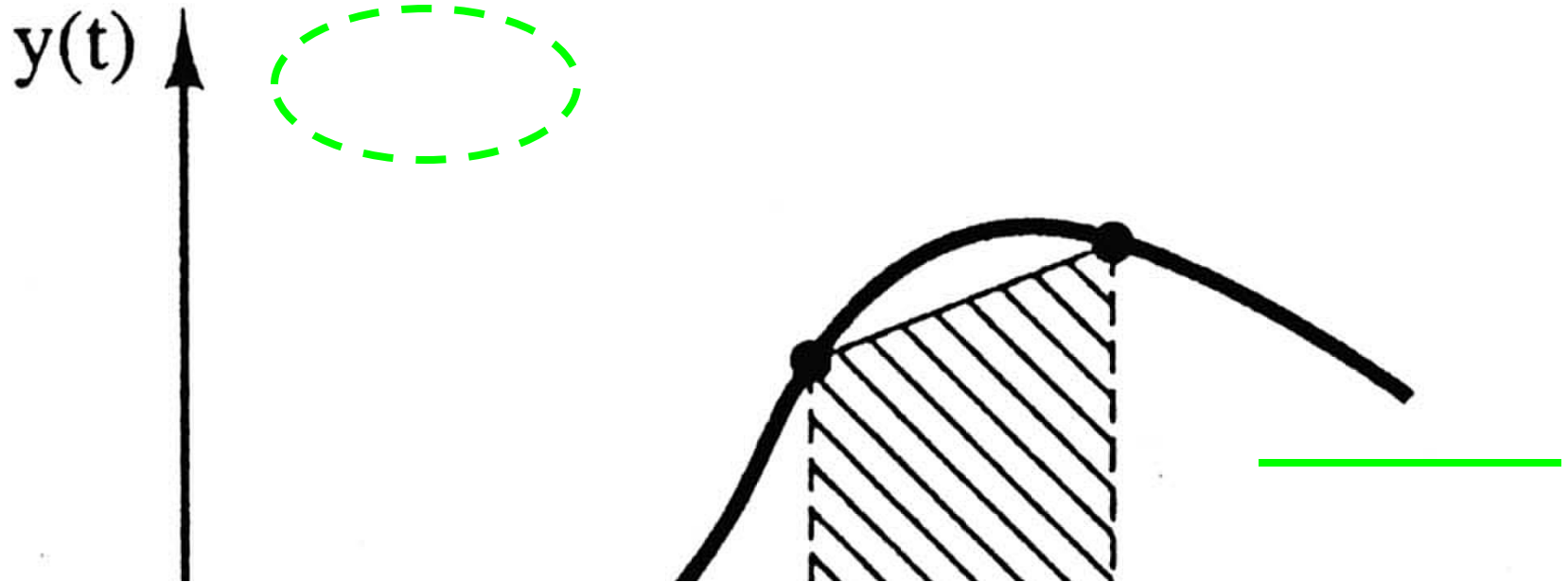
Employing the expressions above for $D(z)$ and $G(z)$, we see that

$$K_v = \lim_{z \rightarrow 1} (z - 1) \frac{(K_I + K_p)z - K_p}{T(z - 1)} \left[\frac{1 - e^{-T}}{z - e^{-T}} \right] = \frac{K_I}{T}$$

Thus K_I equals $100T$ for the required steady-state error, provided that the system is stable. The latter point is indeed an important consideration since the error analysis is

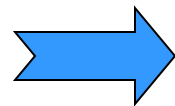
6.6 Simulation





- (P226) 4th-order Runge-Kutta rule
- Two additional points
 - Digital simulation: difference eq.
 - Nonlinear systems

- See Summary (P227)
- Problems: 1, 2, 10, 12,
14, 15, 17

 Test Bank